

PROJECT NAME:

PROPOSED CONSTRUCTION OF IDUTYWA CONFERENCE CENTRE AND ASSOCIATED INFRASTRUCTURE WITHIN THE MBHASHE LOCAL MUNICIPALITY, IN THE EASTERN CAPE

REPORT TITLE:

DRAFT BASIC ASSESSMENT REPORT

DEDEAT REFERENCE NO: EC/9/A/LN1, LN3/M/26-23

DWS REFERENCE: WU51622

DATE: 11 SEPTEMBER 2026

ENVIRONMENTAL ASSESSMENT PRACTITIONER:

ABANTU ENVIRONMENTAL SERVICES (PTY) LTD

PROJECT MANAGERS:

CHIMORA GROUP

**APPLICANT:**

MBHASHE LOCAL MUNICIPALITY



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DATE	July 2026	August 2026	September 2026



BASIC ASSESSMENT REPORT

(For official use only)

File Reference Number:

NEAS Number:

Date Received:

Basic assessment report in terms of the Environmental Impact Assessment Regulations, 2014 as amended, promulgated in terms of the National Environmental Management Act, 1998(Act No. 107 of 1998), as amended.

Kindly note that:

1. This **basic assessment report** is a standard report that may be required by a competent authority in terms of the EIA Regulations, 2014 as amended and is meant to streamline applications. Please make sure that it is the report used by the particular competent authority for the activity that is being applied for. This report is current as **of 4 May 2026**. It is the responsibility of the applicant to ascertain whether subsequent versions of the form have been published or produced by the competent authority
2. The report must be typed within the spaces provided in the form. The size of the spaces provided is not necessarily indicative of the amount of information to be provided. The report is in the form of a table that can extend itself as each space is filled with typing.
3. Where applicable **tick** the boxes that are applicable or **black out** the boxes that are not applicable in the report.
4. An incomplete report may be returned to the applicant for revision.
5. The use of "not applicable" in the report must be done with circumspection because if it is used in respect of material information that is required by the competent authority for assessing the application, it may result in the rejection of the application as provided for in the regulations.
6. This report must be handed in at offices of the relevant competent authority as determined by each authority **unless indicated otherwise by the Department**.
7. No faxed or e-mailed reports will be accepted **unless indicated otherwise by the Department**.
8. The report must be compiled by an independent environmental assessment practitioner (EAP). The EAP must satisfy conditions 11 below.
9. Unless protected by law, all information in the report will become public information on receipt by the competent authority. Any interested and affected party should be provided with the information contained in this report on request, during any stage of the application process.
10. A competent authority may require that for specified types of activities in defined situations only parts of this report need to be completed.
- 11.1 The Environmental Assessment Practitioner (EAP) must be registered in terms of S24H Regulations with the Registration Authority EAPASA as from 8 August 2022.

11.2. S24H (14) states that “only a person registered as an Environmental Assessment practitioner may perform tasks in connection with an application for an environmental authorisation contemplated in

(a) Chapter 5 of the Act read with the Environmental Impact Assessment Regulations.

(b) Section 24G of the Act

(c) Chapter 5 of the National Environmental Management Waste Act 2008 (Act No 59 of 2008) read with the Environmental Impact Assessment Regulations

11.3. Tasks in regulation 14 may only be conducted by an EAP that is registered

11.4. Regulations 20 of S24H indicates the offences and penalties as indicated below:

“20. Offences and penalties

(1) A person is guilty of an offence if that person-

(a) contravenes regulation 14 of the Regulations; or

(b) pretends to be a registered environmental assessment practitioner or registered candidate environmental assessment practitioner.

(2) A person convicted of an offence in terms of subregulation (1) is liable to the penalties contemplated in section 49B(3) of the Act.”.

Section 49B(3) of the Act states:

“A person convicted of an offence in terms of section 49A(1)(h), (l), (m), (n), (o) or (p) is liable to a fine or to imprisonment for a period not exceeding one year, or to both a fine and such imprisonment.”.

SECTION A: ACTIVITY INFORMATION

Has a specialist been consulted to assist with the completion of this section?

YES	
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If YES, please complete form XX for each specialist thus appointed:

Any specialist reports must be contained in Appendix D.

1. ACTIVITY DESCRIPTION

Describe the activity, which is being applied for, in detail

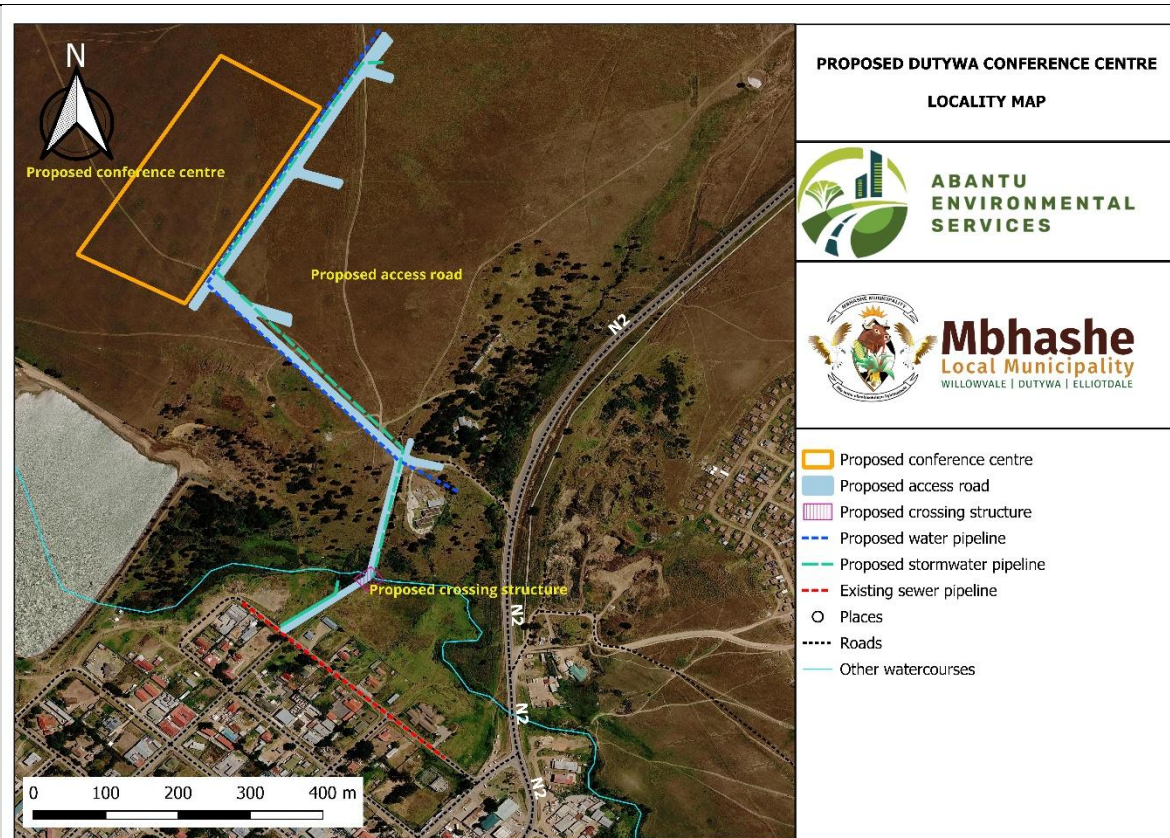
1. Introduction

1.1 Project Location and Description

Abantu Environmental Services (Pty) Ltd has been appointed by Chimora Group (Pty) Ltd on behalf of Mbhashe Local Municipality to apply for an Environmental Authorisation for the proposed construction of Idutywa conference centre and associated infrastructure in Dutywa within the Mbhashe Local Municipality (MLM).

The proposed development comprises of the following:

- Proposed conference centre
- Proposed Sewer Pipe - 160mm Class 34 UPVC and Galvanised Steel Sewer Pipe
- Proposed Water Pipe - main - 8 m servitude
- Proposed Concrete Stormwater Pipes- 450mm – 600mm Ø 100D
- Proposed Sewer Manholes
- Proposed Sw Catchpit/Manhole
- **Proposed concrete/paved/tar access road and a crossing structure (bridge/culvert)**
 - Dimensions**
 - ~1300m (1.3km) length
 - Width of 3700mm x 2 lanes (7.4m + 1.65m = 9.5m)
 - Shoulders including sidewalks;
 - side walks = 1200mm
 - stormwater drains and should of 465mm
 - **Layers works**
 - 60mm thick 35mpa interlocking brick
 - paving over 20mm compacted sand
 - 150mm g5 material stabilised with 3% cement compacted to 98% mod aashto
 - 150mm g7 material compacted to 95% mod 150mm insitu compacted to 93% mod aashtod aashto



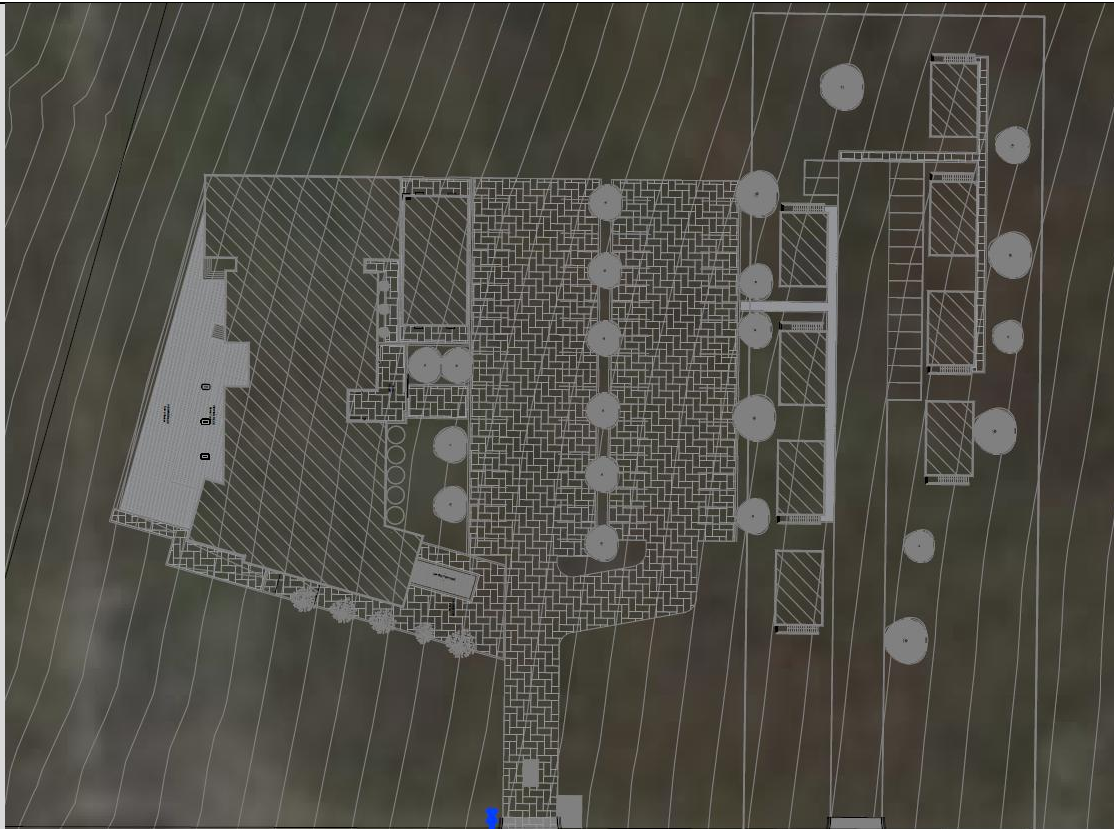


Figure 3. Conference centre layout.

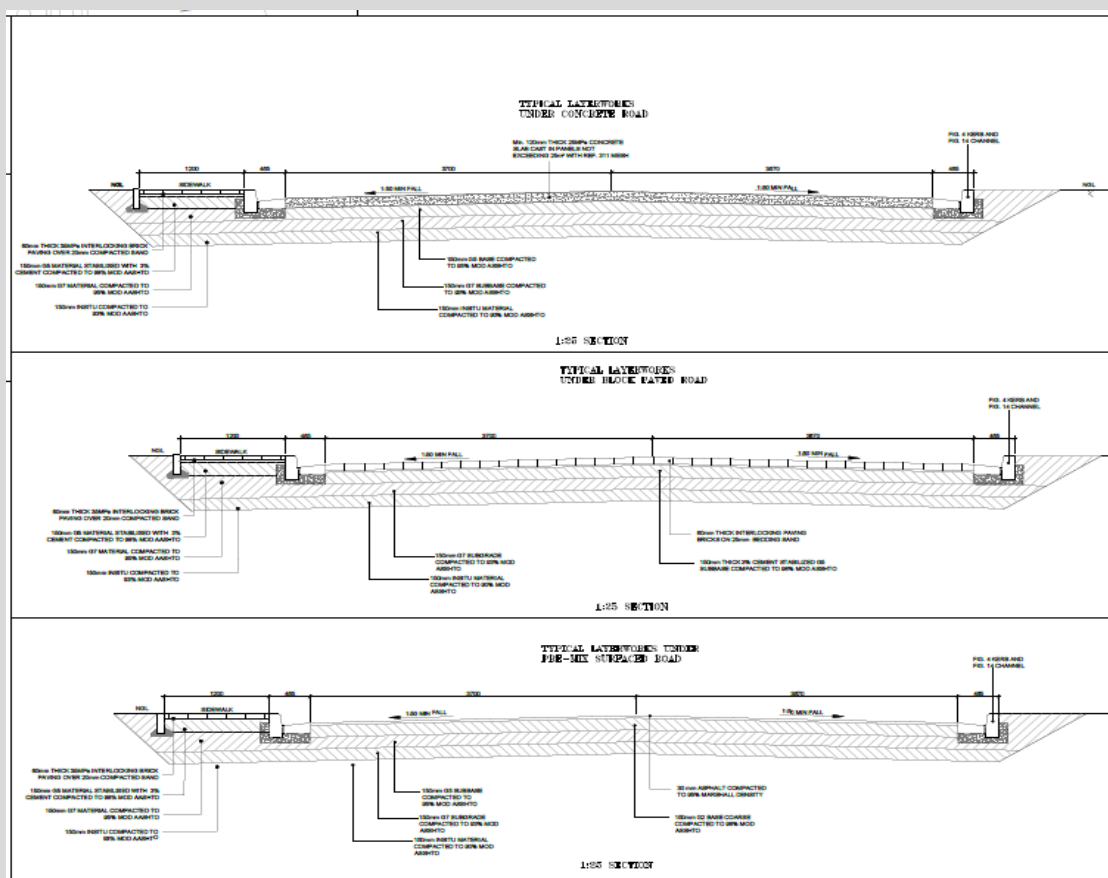


Figure 4. Road cross sections.

2. Biophysical Environment

2.1 Climate

Dutywa is situated at an altitude of approximately 388m above sea level and receives approximately 530 – 540 mm of rainfall annually, with March being the wettest month. The area experiences a warm temperate climate with summer rainfall, characterized by warm summers and cool, dry winters. The warmest conditions are generally experienced during February, with an average midday temperature of about 25°C, while the coldest conditions occur during July, when average temperatures drop to around 18.3°C during the day and 5.1°C at night.

2.2 Topography and hydrology

The area is characterised by undulating to gently rolling inland terrain (**Figure 5**), with the broader municipality displaying fragmented topography of open grasslands, valleys and river corridors. It is generally less rugged than the coastal parts although it is still associated with incised drainage lines, eroded slopes and uneven landforms in places (MLM IDP, 2023).



Figure 5. Elevation profile of the proposed development site.

The area falls within the Mzimvubu-Tsitsikama water management area and within the Mbashe river catchment system. Dutywa is located within the broader Mbashe river basin, where surface water drainage is influenced by a network of perennial and non-perennial streams, smaller drainage lines and associated valley systems. The Mputi river forms the primary watercourse in the immediate vicinity of the project and is traversed by the proposed access road, where the crossing structure will be constructed. The proposed water, sewer and stormwater pipelines will also cross the Mputi river within the same infrastructure corridor (**Figure 7**). The Mputi river flows south-west of the development site, where it joins the Ngxakaxa river (**Figure 8**), and the combined flow subsequently discharges into the Nqabarha river further south, forming part of the broader Mbashe river catchment.

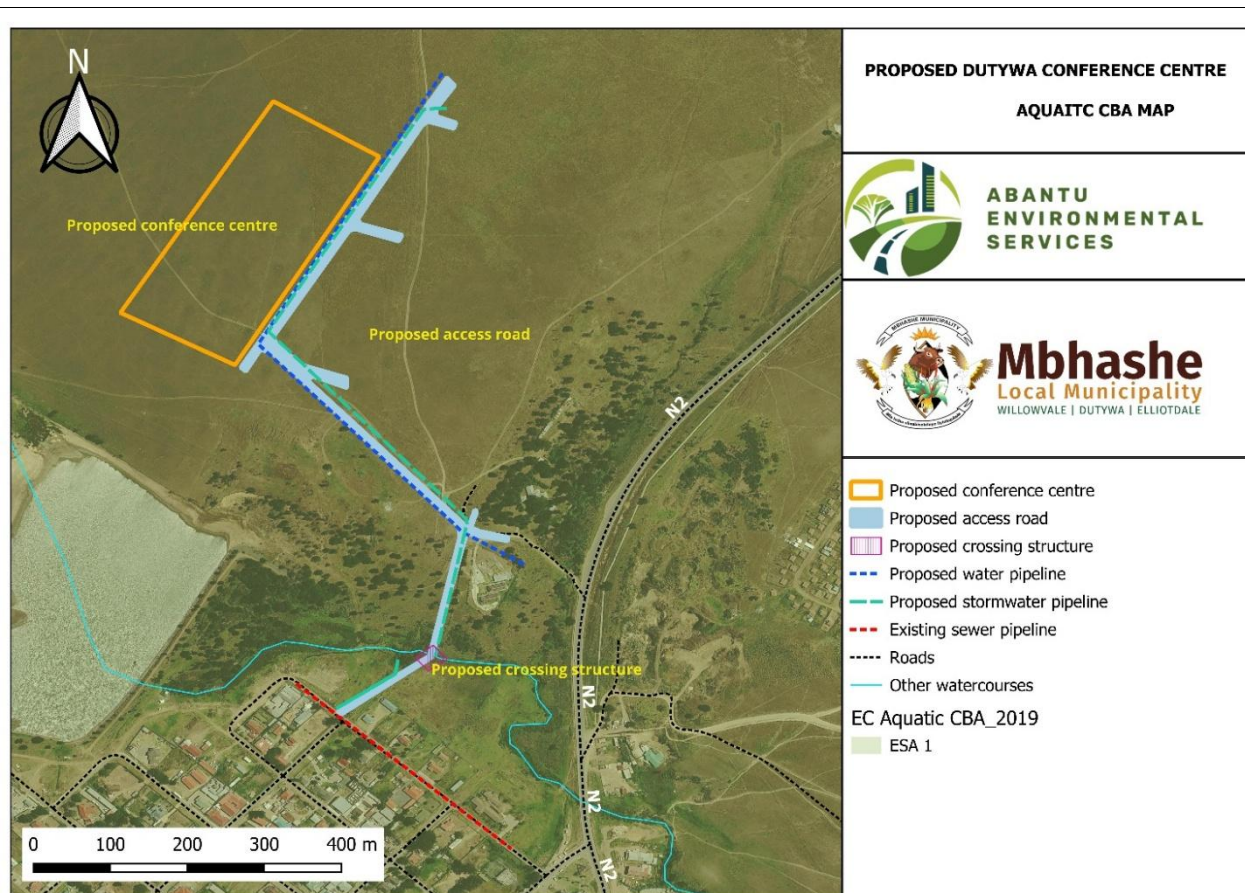


Figure 6. Aquatic biodiversity map of the proposed conference centre and associated infrastructure.



Figure 7. The river that the proposed sewer, water and stormwater pipeline will cross.



Figure 8. The confluence where the Mputi river joins the Ngxakaxa river.

2.3 Soils and Geology

The area is dominated by sedimentary rocks of the Karoo Supergroup, mainly of Late Permian to Early Triassic age, together with younger Jurassic dolerite intrusions. The area is predominantly underlain by the Beaufort Group, particularly the Adelaide Subgroup. The broader Dutywa area is also associated with the Katberg Formation of the Tarkastad Subgroup, which generally comprises of red and greenish-grey mudstone with subordinate fine-to-medium grained sandstone. In some parts of the wider Dutywa area, younger alluvial deposits are also associated with drainage lines and river valleys.

The soils in the Dutywa area are generally derived from the weathering of the underlying mudstone, sandstone and dolerite, and are typically shallow to moderately deep, depending on slope and proximity to drainage features. These soils may be prone to erosion where vegetation has been disturbed or where they occur on steeper slopes, while deeper alluvial soils may occur along valley bottoms.

2.4 Biodiversity

According to SANBI, the proposed conference centre and its associated infrastructure traverses vegetation classified as the Mthatha Moist Grassland (**Figure 9**), which falls under the Grassland Biome. This vegetation type is typically made up of species-poor, sour, wiry grassland with *Eragrostis plana* and *Sporobolus africanus*. The grassland is in good condition, but likely to be dominated by *Themeda triandra* (Mucina et al., 2006). According to the Revised National Ecosystem List (2022), the Mthatha Moist Grassland is listed as Endangered. The proposed development footprint extends the built-up urban area north-eastward

across undeveloped grassland towards the proposed conference centre. Its access road crosses a tributary of Ngxakaxa river, which may support more moisture-dependent vegetation and locally higher ecological sensitive vegetation. According to the Eastern Cape Biodiversity Conservation Plan (2019), the proposed development falls within CBA 2 and ESA 1, indicating areas of ecological sensitivity, although parts of the development footprint are moderately transformed by informal tracks.

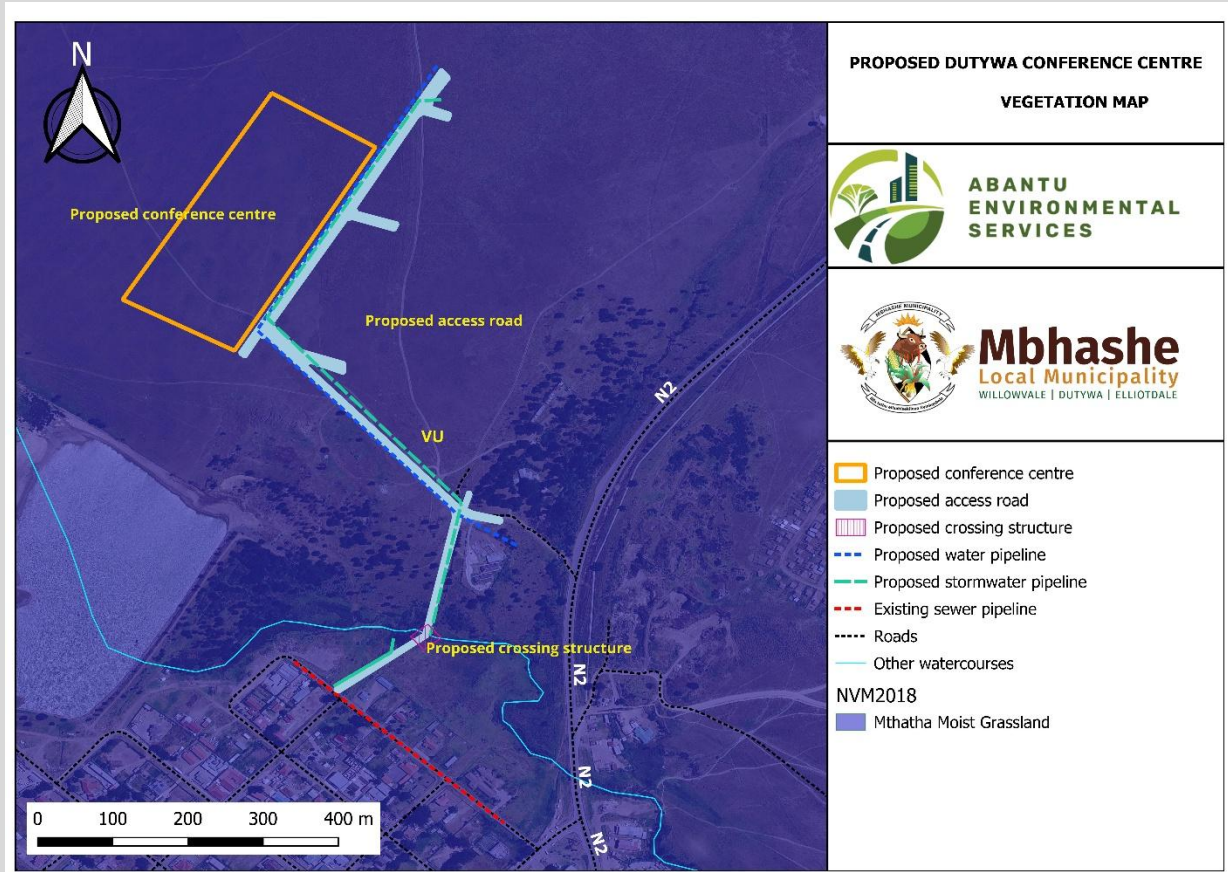


Figure 9. Vegetation map of the proposed conference centre and associated infrastructure.



Figure 10. Vegetation conditions of the proposed conference site.

2.5 Site Sensitivity

The proposed area falls under an Ecological Support Area 1 and Critical Biodiversity Area 2.

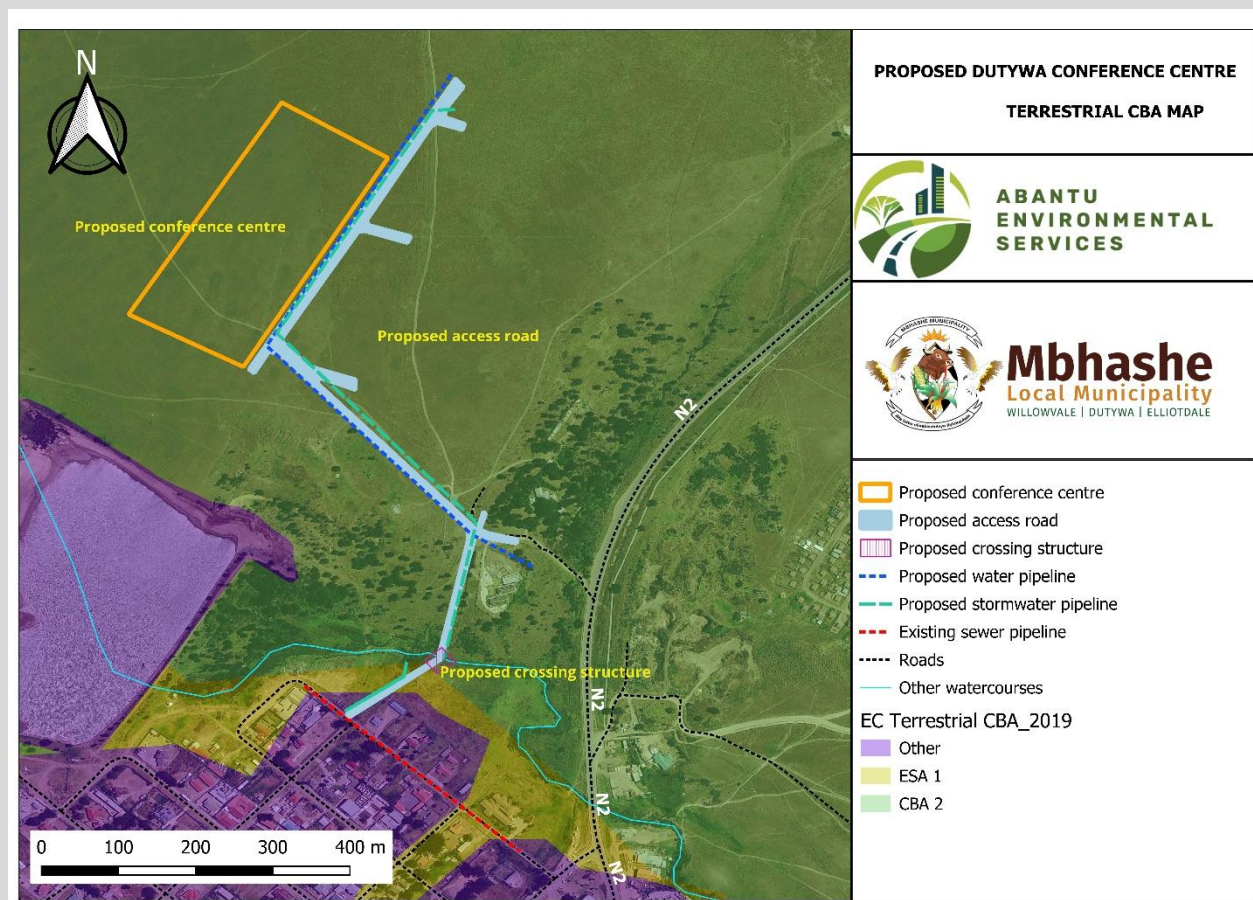


Figure 11. Terrestrial biodiversity map of the proposed conference centre and associated infrastructure.

3. Socio-economic Profile

The proposed project is located in Dutywa within the Mbhashe Local Municipality, in the Eastern Cape Province of South Africa. Mbhashe Local Municipality falls under the Amathole District Municipality and includes the main towns of Dutywa, Willowvale and Elliotdale, together with extensive surrounding rural settlements and agricultural areas. Dutywa serves as one of the key administrative and commercial centres within the municipality. The municipality is characterised by a predominantly rural settlement pattern, with a dispersed population and a network of local and regional roads connecting the towns and surrounding villages. Significant routes in the broader municipal area include the N2 national route, which provides regional connectivity through the Eastern Cape, as well as a number of provincial and local access roads linking Dutywa to nearby settlements and service centres. The municipality is divided into a number of wards that support local governance and service delivery within the municipal area.

3.1 Population

According to Census (2022), the municipality has total population of 240 020 with a household of 52 715 resulting to a household size increase of 4.6% (Census, 2022). When compared to other regions, it accounts for 27.5% of the total population in the district.

Dutywa has a population of approximately 11 076 people (Census 2011), with 54.1% of the population being females and 45.9% being males.

3.2 Household living conditions

Mbhashe Local Municipality has 52 715 households with an average household size of 4.6 persons per household. The majority of residents live in formal dwellings (69.2%). 26.7% of households have access to piped water inside the dwelling, 19.8% have water on community stand, 4.9% access water inside their yards, and 48.6% have no access to piped water. Households that have access to electricity make up 90.5% and 26.5% of households have flush toilets. In terms of waste management, 26.8% households receive weekly refuse removal by local authority (Census, 2022).

3.3 Economic aspect

In 2021, Mbhashe Local Municipality employed a total number of 20 600 people within its local municipality. The community services sector recorded the largest number of employments in 2021 with a total of 7 540 employees. The trade sector with a total of 5 120 employees, the electricity sector with 53 people and the mining sector employing just 26 people. For the period 2011 to 2021 the number of households earning more than R30,000 per annum has increased from 67.61% to 87.50%. The number of households with income equal to or lower than R6,000 per year has decreased by a significant amount (MLM IDP, 2026).

3.4 Education

According to Census 2022, the municipality records that about 5.6% of the population have received Higher Education schooling, 16.5% have Matric, 37% have received Secondary schooling, 6.4% have completed Primary school, and 16% have no schooling.

3.5 Services

3.5.1 Water and Sanitation

Mbhashe Local Municipality is not a Water Service Authority (WSA) nor is a Water Service Provider (WSP), Amathole District Municipality (ADM) is responsible for both WSA & WSP. ADM has WSP satellite units to serve Mbhashe Municipality in each unit and through District Engineering Forum and Water Forum that are facilitated by ADM and there is an integration and information for the planned and implemented projects (MLM IDP, 2026). The proposed conference centre is expected to be supplied with water from existing municipal water distribution network, subject to confirmation of available capacity and the approval of a connection by the municipality. A new water pipeline is proposed to connect the development to the municipal reticulation system. The capacity and pressure of the existing water infrastructure will be confirmed during the detailed design phase to ensure that the conference centre's operational water demand can be adequately met. Should the existing municipal infrastructure be insufficient, appropriate upgrades or on-site water storage facilities may be incorporated into the development.

The proposed development will discharge wastewater into the municipal sewer network via a new sewer pipeline, subject to confirmation that the existing wastewater treatment infrastructure has sufficient capacity to accommodate the additional sewage load. The municipality will confirm the availability of sewer capacity during the detailed design stage. Should the municipal wastewater system be unable to accommodate the proposed development, appropriate engineering solutions, including upgrades to the existing infrastructure or an alternative on-site wastewater treatment system, will be investigated in consultation with the relevant authorities and subject to the necessary environmental and water use authorisations, where applicable.

3.5.2 Electricity

Electricity supply within the Mphashe Local Municipality is provided primarily by Eskom, with the municipality facilitating electricity distribution in certain serviced areas (MLM IDP, 2026). The proposed conference centre will require a reliable electricity supply for its operational requirements. Electricity for the development is expected to be sourced through a new or upgraded connection to the existing Eskom distribution network, subject to an application and approval by Eskom. Existing electricity infrastructure is located in the vicinity of the proposed development site, and the final point of connection and any network upgrades required will be confirmed during the detailed engineering design phase in consultation with Eskom.

4. Local beneficiation

The proposed Dutywa Conference Centre is expected to provide significant socio-economic benefits to the local community through employment creation, local procurement and improved business opportunities. During the construction phase, temporary employment opportunities will be created for local labourers, contractors and suppliers, while the procurement of materials and services from local businesses and SMMEs will stimulate the local economy. Preference should be given to local labour and enterprises where feasible to maximise the project's economic benefits.

During the operational phase, the conference centre is expected to create permanent employment opportunities in administration, hospitality, catering, security, maintenance and event management. The facility will provide a venue for conferences, training programmes, government meetings, and community events, while attracting visitors to Dutywa and contributing to long-term economic growth and development within the Mphashe Local Municipality.

2. FEASIBLE AND REASONABLE ALTERNATIVES

“alternatives”, in relation to a proposed activity, means different means of meeting the general purpose and requirements of the activity, which may include alternatives to—

- (a) the property on which or location where it is proposed to undertake the activity;
- (b) the type of activity to be undertaken;
- (c) the design or layout of the activity;
- (d) the technology to be used in the activity;
- (e) the operational aspects of the activity; and
- (f) the option of not implementing the activity.

Describe alternatives that are considered in this application. Alternatives should include a consideration of all possible means by which the purpose and need of the proposed activity could be accomplished in the specific instance taking account of the interest of the applicant in the activity. The no-go alternative must in all cases be included in the assessment phase as the baseline against which the impacts of the other alternatives are assessed. The determination of whether site or activity (including different processes etc.) or both is appropriate needs to be informed by the specific circumstances of the activity and its environment. After receipt of this report the competent authority may also request the applicant to assess additional

alternatives that could possibly accomplish the purpose and need of the proposed activity if it is clear that realistic alternatives have not been considered to a reasonable extent.

Paragraphs 3 – 13 below should be completed for each alternative.

a) Site alternatives
Site Alternative 1 (proposed) This site represents a compact development footprint located on the north-eastern outskirts of Dutywa immediately adjacent to existing municipal infrastructure. The proposed development with its associated infrastructure has been strategically located to maximise accessibility while maintaining close proximity to the existing road network and surrounding residential and commercial areas. This alternative benefits from improved access to existing municipal services and connectivity to the Dutywa central business district. Although the development will extend into an undeveloped grassland area and require the crossing of the Mputi River, the proposed layout consolidates the development footprint within a single corridor, thereby reducing unnecessary land disturbance and limiting the fragmentation of surrounding natural areas. There is no other site alternative for the proposed development. The property has been strategically reserved for the proposed development and its associated infrastructure. Layout alternatives Layout alternatives for the proposed Dutywa conference centre and associated infrastructure were informed by the two identified site alternatives. Layout Alternative 1 This alternative confines the conference centre and associated infrastructure within a single development corridor. The water, sewer and stormwater pipelines follow the same alignment as the proposed access road, thereby minimizing the extent of linear infrastructure and reducing the duplication of service corridors. This layout limits the fragmentation of the development footprint by concentrating the associated infrastructure within one area. Layout Alternative 2 This alternative proposes locating the access route along the existing gravel road that connects directly to the N2, thereby avoiding the need to construct a crossing over the Mputi river. By making use of an existing access corridor, this alternative would reduce direct impacts on the river and its associated riparian habitat. However, the suitability of the existing gravel road would need to be assessed in terms of traffic capacity, engineering requirements, and the availability of municipal services. Notwithstanding, from a terrestrial and aquatic biodiversity lenses, this alternative is preferred. Moreover, from a social view, in particular accidents and safety as the access road from N2 is a potentially high risk impact. Thus, the first alternative although it will introduce ecological impacts, they are prominently limited to during construction.

b) Technology alternative

No feasible technology alternatives have been identified for the proposed development. The conference centre will be connected to the existing Eskom electricity distribution network, subject to confirmation of network capacity and approval of a new electrical connection. To improve energy efficiency and enhance resilience during power interruptions, the incorporation of renewable energy technologies, such as rooftop solar photovoltaic (PV) panels with battery storage and a standby generator, may be considered during the detailed design phase. These technologies would reduce reliance on the national grid, improve energy security and contribute to more sustainable operation of the facility.

The proposed development will obtain potable water from the municipal water supply network and discharge wastewater to the municipal sewer system via new service connections. The capacity of the existing water and sewer infrastructure will be confirmed by the Mphashe Local Municipality during the detailed design stage. If the available water pressure or storage capacity is found to be inadequate, on-site water storage tanks may be incorporated into the design to ensure a reliable water supply. Similarly, should the municipal sewer network or wastewater treatment works have insufficient capacity to accommodate the additional wastewater load, appropriate engineering solutions, including upgrades to the municipal infrastructure or a suitable on-site wastewater treatment system, will be investigated in consultation with the relevant authorities and subject to the necessary approvals.

c) Operational alternative

No other operational alternatives were considered feasible, as the proposed operation best aligns with the identified need for a multi-purpose conference and events facility in the area.

d) No-Go alternative

The No-Go alternative entails not proceeding with the development of the proposed Dutywa Conference Centre and its associated infrastructure. Under this alternative, the site would remain in its current undeveloped state, thereby avoiding the direct environmental impacts associated with construction, such as vegetation clearance, disturbance of the Mputi river crossing, and temporary impacts on terrestrial and aquatic ecosystems. However, the existing lack of dedicated conference and events facilities within Dutywa would remain unaddressed.

Implementing the No-Go alternative would also forgo the anticipated socio-economic benefits of the proposed development, including employment creation during the construction and operational phases, increased opportunities for local businesses and SMMEs, enhanced tourism and business activities, and the provision of a modern venue for conferences, government meetings, training programmes, and community events. Consequently, while the No-Go Alternative would avoid environmental disturbance in the short term, it would not contribute to the local economic development objectives of the Mphashe Local Municipality or address the identified need for improved conferencing and event infrastructure. As such, the No-Go Alternative is not considered the preferred option.

3. ACTIVITY POSITION

Indicate the position of the activity using the latitude and longitude of the centre point of the site for each alternative site. The co-ordinates should be in degrees and decimal minutes. The minutes should have at least three decimals to ensure adequate accuracy. The projection that must be used in all cases is the WGS84 spheroid in a national or local projection.

List alternative sites if applicable.

	Latitude (S):	Longitude (E):
Alternative:		
Alternative S1 ¹ (preferred or only site alternative)	See Appendix G	
Alternative S2 (if any)		
Alternative S3 (if any)		

In the case of linear activities:

	Latitude (S):	Longitude (E):
Alternative:		
Alternative S1 (preferred or only route alternative)		
• Starting point of the activity		
• Middle point of the activity		
• End point of the activity		
Alternative S2 (if any)		
• Starting point of the activity		
• Middle point of the activity		
• End point of the activity		
Alternative S3 (if any)		
• Starting point of the activity		
• Middle point of the activity		
• End point of the activity		

For route alternatives that are longer than 500m, please provide an addendum with co-ordinates taken every 250 meters along the route for each alternative alignment.

4. PHYSICAL SIZE OF THE ACTIVITY

Indicate the physical size of the preferred activity/technology as well as alternative activities/technologies (footprints):

Alternative:	Size of the activity:
Alternative A1 ² (preferred activity alternative)	±1 ha
Alternative A2 (if any)	
Alternative A3 (if any)	

¹ "Alternative S.." refer to site alternatives.

² "Alternative A.." refer to activity, process, technology or other alternatives.

or, for linear activities:

- Alternative:**
Alternative A1 (preferred activity alternative)
Alternative A2 (if any)
Alternative A3 (if any)

Length of the activity:

Indicate the size of the alternative sites or servitudes (within which the above footprints will occur):

- Alternative:**
Alternative A1 (preferred activity alternative)
Alternative A2 (if any)
Alternative A3 (if any)

Size of the site/servitude:

5. SITE ACCESS

Does ready access to the site exist?
If NO, what is the distance over which a new access road will be built

No

Describe the type of access road planned:

The proposed conference centre is boarded by existing municipal roads and undeveloped land within the urban edge of Dutywa. The site is accessed from the existing public road network via a proposed access road that extends from the surrounding municipal road system toward the development footprint. It is proposed that vehicular and pedestrian access to the site will be provided from this new access road, which will include a crossing structure over the Mputi river to provide safe and reliable access to the development.



Figure 12. Site access to the proposed site.

Include the position of the access road on the site plan and required map, as well as an indication of the road in relation to the site.

6. SITE OR ROUTE PLAN

A detailed site or route plan(s) must be prepared for each alternative site or alternative activity. It must be attached as Appendix A to this document.

The site or route plans must indicate the following:

- 6.1 the scale of the plan which must be at least a scale of 1:500;
- 6.2 the property boundaries and numbers of all the properties within 50 metres of the site;
- 6.3 the current land use as well as the land use zoning of each of the properties adjoining the site or sites;
- 6.4 the exact position of each element of the application as well as any other structures on the site;
- 6.5 the position of services, including electricity supply cables (indicate above or underground), water supply pipelines, boreholes, street lights, sewage pipelines, storm water infrastructure and telecommunication infrastructure;
- 6.6 all trees and shrubs taller than 1.8 metres;
- 6.7 walls and fencing including details of the height and construction material;
- 6.8 servitudes indicating the purpose of the servitude;
- 6.9 sensitive environmental elements within 100 metres of the site or sites including (but not limited thereto):
 - rivers;
 - the 1:100 year flood line (where available or where it is required by DWA);
 - ridges;
 - cultural and historical features;
 - areas with indigenous vegetation (even if it is degraded or invested with alien species);
- 6.9 for gentle slopes the 1 metre contour intervals must be indicated on the plan and whenever the slope of the site exceeds 1:10, the 500mm contours must be indicated on the plan; and
- 6.10 the positions from where photographs of the site were taken.

7. SITE PHOTOGRAPHS

Colour photographs from the centre of the site must be taken in at least the eight major compass directions with a description of each photograph. Photographs must be attached under Appendix B to this form. It must be supplemented with additional photographs of relevant features on the site, if applicable.

8. FACILITY ILLUSTRATION

A detailed illustration of the activity must be provided at a scale of 1:200 as Appendix C for activities that include structures. The illustrations must be to scale and must represent a realistic image of the planned activity. The illustration must give a representative view of the activity.

9. ACTIVITY MOTIVATION

9(a) Socio-economic value of the activity

What is the expected capital value of the activity on completion?	R75 million
What is the expected yearly income that will be generated by or as a result of the activity?	R10 million
Will the activity contribute to service infrastructure?	YES
Is the activity a public amenity?	YES
How many new employment opportunities will be created in the development phase of the activity?	TBC
What is the expected value of the employment opportunities during the development phase?	150
What percentage of this will accrue to previously disadvantaged individuals?	90%
How many permanent new employment opportunities will be created during the operational phase of the activity?	TBC
What is the expected current value of the employment opportunities during the first 10 years?	R45 million
What percentage of this will accrue to previously disadvantaged individuals?	90%

9(b) Need and desirability of the activity

Motivate and explain the need and desirability of the activity (including demand for the activity):

The proposed development is intended to address the growing need for a dedicated, modern venue capable of accommodating conferences, government meetings, training programmes, exhibition, cultural and community events within the Mbhashe Local Municipality. At present, Dutywa has limited formal conferencing facilities capable of hosting large-scale events, resulting in government departments, businesses, educational institutions and community organisations relying on smaller venues or travelling to neighbouring towns to access suitable facilities. The proposed conference centre will provide a centrally located, purpose-built facility that will improve service delivery, enhance access to meeting and event infrastructure, and support the municipality's social and economic development objectives.

The project is expected to contribute to local economic development by creating employment opportunities during both the construction and operational phases, while increasing demand for local accommodation establishments, restaurants, transport operators, catering services and other SMMEs. The facility will serve as a catalyst for business tourism and investment by attracting conferences, seminars and regional events to Dutywa, thereby generating income for the local economy. Additionally, the project includes supporting infrastructure such as an access road, crossing structure over the Mputi river and municipal service connections, which will improve accessibility and service provision within the area.

Indicate any benefits that the activity will have for society in general:

- The proposed development will improve service delivery by enabling government departments to host outreach programmes, workshops and community consultations locally. It supports knowledge sharing and lifelong learning through conferences, workshops and educational institutions and civil society organisations. Extends municipal water, sewer and stormwater services to the development area, which may facilitate future development in the vicinity. Moreover, the stimulation of the local economy through increased demand for local contractors, suppliers, accommodation establishments, catering services and other SMMEs.

Indicate any benefits that the activity will have for the local communities where the activity will be located:

The benefits of the proposed project are that:

- Creation of temporary and permanent employment for the residents during the construction and operational phases.
- Increased business opportunities for local contractors, suppliers, SMMEs and informal traders through the procurement of goods and services.
- Skills development and capacity building through employment, training opportunities and the hosting of educational and professional development events.
- Improvement of local infrastructure through the construction of the access road, Mputi river crossing structure and associated municipal water, sewer and stormwater services.
- Enhancement of community development by providing a shared public facility that supports social interaction, collaboration and inclusive participation in community and economic activities.

10. APPLICABLE LEGISLATION, POLICIES AND/OR GUIDELINES

List all legislation, policies and/or guidelines of any sphere of government that are applicable to the application as contemplated in the EIA regulations, if applicable:

Title of legislation, policy or guideline:	Administering authority:	Date:
GN R.327 Activity 12 (ii) The development of— (i) dams or weirs, where the dam or weir, including infrastructure and water surface area, exceeds 100 square metres; or (ii) infrastructure or structures with a physical footprint of 100 square metres or more; where such development occurs— (a) within a watercourse; (b) in front of a development setback; or (c) if no development setback exists, within 32 metres of a watercourse, measured from the edge of a watercourse; — excluding— (aa) the development of infrastructure or structures within existing ports or harbours that will not increase the development footprint of the port or harbour; (bb) where such development activities are related to the development of a port or harbour, in which case activity 26 in Listing Notice 2 of 2014 applies; (cc) activities listed in activity 14 in Listing Notice 2 of 2014 or activity 14 in Listing Notice 3 of 2014, in which case that activity applies; (dd) where such development occurs within an urban area; (ee) where such development occurs within existing roads, road reserves or railway line reserves; or	DEDEAT	December 2017

Title of legislation, policy or guideline:**Administering authority:****Date:**

(ff) the development of temporary infrastructure or structures where such infrastructure or structures will be removed within 6 weeks of the commencement of development and where indigenous vegetation will not be cleared.		
GN R.327 Activity 19 The infilling or depositing of any material of more than 10 cubic metres into, or the dredging, excavation, removal or moving of soil, sand, shells, shell grit, pebbles or rock of more than 10 cubic metres from but excluding where such infilling, depositing, dredging, excavation, removal or moving— (a) will occur behind a development setback; (b) is for maintenance purposes undertaken in accordance with a maintenance management plan; [or] (c) falls within the ambit of activity 21 in this Notice, in which case that activity applies; (d) occurs within existing ports or harbours that will not increase the development footprint of the port or harbour; or (e) where such development is related to the development of a port or harbour, in which case activity 26 in Listing Notice 2 of 2014 applies.	DEDEAT	December 2017
GN R.327 Activity 24 (ii) The development of a road— (i) for which an environmental authorisation was obtained for the route determination in terms of activity 5 in Government Notice 387 of 2006 or activity 18 in Government Notice 545 of 2010; or (ii) with a reserve wider than 13,5 meters, or where no reserve exists where the road is wider than 8 metres; but excluding a road— (a) which is identified and included in activity 27 in Listing Notice 2 of 2014; (b) where the entire road falls within an urban area; or (c) which is 1 kilometre or shorter.	DEDEAT	December 2017
GNR 327 27 The clearance of an area of 1 hectares or more, but less than 20 hectares of indigenous vegetation, except where such clearance of indigenous vegetation is required for— (i) the undertaking of a linear activity; or (ii) maintenance purposes undertaken in accordance with a maintenance management plan.	DEDEAT	December 2017
GNR 327 28 Residential, mixed, retail, commercial, industrial or institutional developments where such land was used for agriculture, game farming, equestrian purposes or afforestation on or after 01 April 1998 and where such development: (i) will occur inside an urban area, where the total land to be developed is bigger than 5 hectares; or (ii) will occur outside an urban area, where the total land to be developed is bigger than 1 hectare; excluding where such land has already been developed for residential, mixed, retail, commercial, industrial or institutional purposes.	DEDEAT	December 2017
GN R.324 Activity 4. a.i. (ee)	DEDEAT	December 2017

Title of legislation, policy or guideline:

Administering authority: **Date:**

The development of a road wider than 4 metres with a reserve less than 13,5 metres. a. Eastern Cape i. Outside urban areas: (aa) A protected area identified in terms of NEMPAA, excluding disturbed areas; (bb) National Protected Area Expansion Strategy Focus areas; (cc) Sensitive areas as identified in an environmental management framework as contemplated in chapter 5 of the Act and as adopted by the competent authority; (dd) Sites or areas identified in terms of an international convention; (ee) Critical biodiversity areas as identified in systematic biodiversity plans adopted by the competent authority or in bioregional plans; (ff) Core areas in biosphere reserves; (gg) Areas within 10 kilometres from national parks or world heritage sites or 5 kilometres from any other protected area identified in terms of NEMPAA or from the core areas of a biosphere reserve, excluding disturbed areas; (hh) Areas seawards of the development setback line or within 1 kilometre from the high-water mark of the sea if no such development setback line is determined; or (ii) In an estuarine functional zone, excluding areas falling behind the development setback line; or ii. Inside urban areas: (aa) Areas zoned for use as public open space; (bb) Areas designated for conservation use in Spatial Development Frameworks adopted by the competent authority or zoned for a conservation purpose; or (cc) Seawards of the development setback line or within urban protected areas.		
GNR 324 12 a.i. (i)(v) The clearance of an area of 300 square metres or more of indigenous vegetation except where such clearance of indigenous vegetation is required for maintenance purposes undertaken in accordance with a maintenance management plan. a. Eastern Cape i. Within any critically endangered or endangered ecosystem listed in terms of section 52 of the NEMBA or prior to the publication of such a list, within an area that has been identified as critically endangered in the National Spatial Biodiversity Assessment 2004; ii. Within critical biodiversity areas identified in bioregional plans; iii. Within the littoral active zone or 100 metres inland from the high water mark of the sea, whichever distance is the greater, excluding where such removal will occur behind the development setback line on erven in urban areas; iv. Outside urban areas, within 100 metres inland from an estuarine functional zone; or v. On land, where, at the time of the coming into effect of this Notice or thereafter such land was zoned open space, conservation or had an equivalent zoning.	DEDEAT	December 2017
GN R.324 Activity 14 (ii).a.i.(ff) The development of— (i) dams or weirs, where the dam or weir, including infrastructure and water surface area exceeds 10 square metres; or (ii) infrastructure or structures with a physical footprint of 10 square metres or more; where such development occurs— (a) within a watercourse; (b) in front of a development setback; or	DEDEAT	December 2017

Title of legislation, policy or guideline:

Administering authority: **Date:**

(c) if no development setback has been adopted, within 32 metres of a watercourse, measured from the edge of a watercourse; excluding the development of infrastructure or structures within existing ports or harbours that will not increase the development footprint of the port or harbour. a. Eastern Cape i. Outside urban areas: (aa) A protected area identified in terms of NEMPAA, excluding conservancies; (bb) National Protected Area Expansion Strategy Focus areas; (cc) World Heritage Sites; (dd) Sensitive areas as identified in an environmental management framework as contemplated in chapter 5 of the Act and as adopted by the competent authority; (ee) Sites or areas identified in terms of an international convention; (ff) Critical biodiversity areas or ecosystem service areas as identified in systematic biodiversity plans adopted by the competent authority or in bioregional plans; (gg) Core areas in biosphere reserves; (hh) Areas within 10 kilometres from national parks or world heritage sites or 5 kilometres from any other protected area identified in terms of NEMPAA or from the core area of a biosphere reserve; (ii) Areas seawards of the development setback line or within 1 kilometre from the high-water mark of the sea if no such development setback line is determined; or (jj) In an estuarine functional zone, excluding areas falling behind the development setback line; or ii. Inside urban areas: (aa) Areas zoned for use as public open space; (bb) Areas designated for conservation use in Spatial Development Frameworks adopted by the competent authority, zoned for a conservation purpose; or (cc) Areas seawards of the development setback line.		
GN R.324 Activity 15 a.i. The transformation of land bigger than 1000 square metres in size, to residential, retail, commercial, industrial or institutional use, where, such land was zoned open space, conservation or had an equivalent zoning, on or after 02 August 2010.	DEDEAT	December 2017

11. WASTE, EFFLUENT, EMISSION AND NOISE MANAGEMENT

11(a) Solid waste management

Will the activity produce solid construction waste during the construction/initiation phase?

If yes, what estimated quantity will be produced per month?

How will the construction solid waste be disposed of (describe)?

YES	
Unknown	

Construction waste will be hauled away by trucks and taken to the nearest Dutywa registered landfill site. Manifests or a waste disposal register will be kept as proof of safe disposal.

Where will the construction solid waste be disposed of (describe)?

Construction waste will be disposed of at the nearest registered landfill site.

Will the activity produce solid waste during its operational phase?

YES

If yes, what estimated quantity will be produced per month?

Unknown

How will the solid waste be disposed of (describe)?

During operational phase general waste such as litter will be generated by the guests at the conference centre. This waste will be collected and stored in an appropriate area, and will be collected by the Mbhashe Local Municipality and disposed of at the landfill site.

Where will the solid waste be disposed if it does not feed into a municipal waste stream (describe)?

Waste from the conference centre will be collected by the municipality and disposed at the landfill site.

If the solid waste (construction or operational phases) will not be disposed of in a registered landfill site or be taken up in a municipal waste stream, then the applicant should consult with the competent authority to determine whether it is necessary to change to an application for scoping and EIA.

Can any part of the solid waste be classified as hazardous in terms of the relevant legislation?

NO

If yes, inform the competent authority and request a change to an application for scoping and EIA.

Is the activity that is being applied for a solid waste handling or treatment facility?

NO

If yes, then the applicant should consult with the competent authority to determine whether it is necessary to change to an application for scoping and EIA.

11(b) Liquid effluent

Will the activity produce effluent, other than normal sewage, that will be disposed of in a municipal sewage system?

NO

If yes, what estimated quantity will be produced per month?

N/A

Will the activity produce any effluent that will be treated and/or disposed of on site?

NO

If yes, the applicant should consult with the competent authority to determine whether it is necessary to change to an application for scoping and EIA.

Will the activity produce effluent that will be treated and/or disposed of at another facility?

NO

If yes, provide the particulars of the facility:

Facility name:	N/A			
Contact person:				
Postal address:				
Postal code:				
Telephone:				Cell:
E-mail:				Fax:

Describe the measures that will be taken to ensure the optimal reuse or recycling of waste water, if any:

N/A

11(c) Emissions into the atmosphere

Will the activity release emissions into the atmosphere?

YES	
	NO

If yes, is it controlled by any legislation of any sphere of government?

If yes, the applicant should consult with the competent authority to determine whether it is necessary to change to an application for scoping and EIA.

If no, describe the emissions in terms of type and concentration:

Construction vehicle emissions and dust during construction phase. Sources of emissions include those produced by heavy diesel vehicles involved in site preparation of the proposed building. These activities will occur only over a short period.
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11(d) Generation of noise

Will the activity generate noise?

YES	
	NO

If yes, is it controlled by any legislation of any sphere of government?

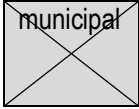
If yes, the applicant should consult with the competent authority to determine whether it is necessary to change to an application for scoping and EIA.

If no, describe the noise in terms of type and level:

Construction activity generated noise and noise levels. Noise will only be generated during construction phase due to construction vehicles and general construction activities.
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12. WATER USE

Please indicate the source(s) of water that will be used for the activity by ticking the appropriate box(es)



If water is to be extracted from groundwater, river, stream, dam, lake or any other natural feature, please indicate the volume that will be extracted per month:

Does the activity require a water use permit from the Department of Water Affairs?



If yes, please submit the necessary application to the Department of Water Affairs and attach proof thereof to this application if it has been submitted.

13. ENERGY EFFICIENCY

Describe the design measures, if any, that have been taken to ensure that the activity is energy efficient:

To be confirmed.

Describe how alternative energy sources have been taken into account or been built into the design of the activity, if any:

The proposed development can incorporate alternative energy sources into the design to reduce reliance on conventional electricity and lower the building's environmental impact. While no large-scale alternative energy infrastructure is confirmed at this stage, the proposed development can be positioned or designed to make maximum use of natural light and ventilation, reducing the amount of electricity required for lighting and air conditioning. The conference centre can make use of energy-efficient equipment such as LED lighting, efficient heating and cooling systems, and energy-saving appliances to reduce the centre's overall energy demand.

SECTION B: SITE/AREA/PROPERTY DESCRIPTION

Important notes:

1. For linear activities (pipelines, etc) as well as activities that cover very large sites, it may be necessary to complete this section for each part of the site that has a significantly different environment. In such cases please complete copies of Section C and indicate the area, which is covered by each copy No. on the Site Plan.

Section C Copy No. (e.g. A):

2. Paragraphs 1 - 6 below must be completed for each alternative.

3. Has a specialist been consulted to assist with the completion of this section?

YES	
-----	--

If YES, please complete form XX for each specialist thus appointed:

All specialist reports must be contained in Appendix D.

1. GRADIENT OF THE SITE

Indicate the general gradient of the site.

Alternative S1:

Flat

Alternative S2 (if any):

Alternative S3 (if any):

2. LOCATION IN LANDSCAPE

Indicate the landform(s) that best describes the site:

- 2.1 Ridgeline
- 2.2 Plateau
- 2.3 Side slope of hill/mountain
- 2.4 Closed valley
- 2.5 Open valley
- 2.6 Plain
- 2.7 Undulating plain / low hills
- 2.8 Dune
- 2.9 Seafront

3. GROUNDWATER, SOIL AND GEOLOGICAL STABILITY OF THE SITE

Is the site(s) located on any of the following (tick the appropriate boxes)?

Alternative S1:

Alternative S2 (if any):

Alternative S3 (if any):

Shallow water table (less than 1.5m deep)

	NO
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Dolomite, sinkhole or doline areas

NO

Seasonally wet soils (often close to water bodies)

NO

Unstable rocky slopes or steep slopes with loose soil

NO

Dispersive soils (soils that dissolve in water)

NO

Soils with high clay content (clay fraction more than 40%)

NO

Any other unstable soil or geological feature

NO

An area sensitive to erosion

NO

If you are unsure about any of the above or if you are concerned that any of the above aspects may be an issue of concern in the application, an appropriate specialist should be appointed to assist in the completion of this section. (Information in respect of the above will often be available as part of the project information or at the planning sections of local authorities. Where it exists, the 1:50 000 scale Regional Geotechnical Maps prepared by the Council for Geo Science may also be consulted).

4. GROUNDCOVER

Indicate the types of groundcover present on the site:

4.1 Natural veld – good condition ^E

4.2 Natural veld – scattered aliens ^E

4.3 Natural veld with heavy alien infestation ^E

4.4 Veld dominated by alien species ^E

4.5 Gardens

4.6 Sport field

4.7 Cultivated land

4.8 Paved surface

4.9 Building or other structure

4.10 Bare soil

The location of all identified rare or endangered species or other elements should be accurately indicated on the site plan(s).

Natural veld with
scattered aliens^E

If any of the boxes marked with an “E” is ticked, please consult an appropriate specialist to assist in the completion of this section if the environmental assessment practitioner doesn’t have the necessary expertise.

5. LAND USE CHARACTER OF SURROUNDING AREA

Indicate land uses and/or prominent features that currently occur within a 500m radius of the site and give description of how this influences the application or may be impacted upon by the application:

- 5.1 Natural area
- 5.2 Low density residential
- 5.3 Medium density residential
- 5.4 High density residential
- 5.5 Informal residential
- 5.6 Retail commercial & warehousing
- 5.7 Light industrial
- 5.8 Medium industrial ^{AN}
- 5.9 Heavy industrial ^{AN}
- 5.10 Power station
- 5.11 Office/consulting room
- 5.12 Military or police base/station/compound
- 5.13 Spoil heap or slimes dam^A
- 5.14 Quarry, sand or borrow pit
- 5.15 Dam or reservoir
- 5.16 Hospital/medical centre
- 5.17 School
- 5.18 Tertiary education facility
- 5.19 Church
- 5.20 Old age home
- 5.21 Sewage treatment plant^A
- 5.22 Train station or shunting yard ^N
- 5.23 Railway line ^N
- 5.24 Major road (4 lanes or more) ^N
- 5.25 Airport ^N
- 5.26 Harbour
- 5.27 Sport facilities
- 5.28 Golf course
- 5.29 Polo fields
- 5.30 Filling station ^H

- 5.31 Landfill or waste treatment site
- 5.32 Plantation
- 5.33 Agriculture
- 5.34 River, stream or wetland
- 5.35 Nature conservation area
- 5.36 Mountain, koppie or ridge
- 5.37 Museum
- 5.38 Historical building
- 5.39 Protected Area
- 5.40 Graveyard
- 5.41 Archaeological site
- 5.42 Other land uses (describe)

If any of the boxes marked with an "N" are ticked, how will this impact / be impacted upon by the proposed activity.
 Not applicable

If any of the boxes marked with an "An" are ticked, how will this impact / be impacted upon by the proposed activity.
 If YES, specify and explain:

Not applicable

If YES, specify:

Not applicable

If any of the boxes marked with an "H" are ticked, how will this impact / be impacted upon by the proposed activity.
 If YES, specify and explain:

Not applicable

If YES, specify:

Not applicable

6. CULTURAL/HISTORICAL FEATURES

Are there any signs of culturally or historically significant elements, as defined in section 2 of the National Heritage Resources Act, 1999, (Act No. 25 of 1999), including Archaeological or palaeontological sites, on or close (within 20m) to the site?	NO The proposed site has a low heritage sensitivity. However, a Notice of Intent to Develop (NID) under Section 38 of the NHRA will be submitted to ECPHRA to determine whether an assessment is required or not.
If YES, explain:	
If uncertain, conduct a specialist investigation by a recognised specialist in the field to establish whether there is such a feature(s) present on or close to the site.	
Briefly explain the findings of the specialist:	No. There is no evidence or archaeological excavations, surveys, or mitigation and no archaeological or heritage remains have been identified in the development area based on available satellite imagery and maps (Umsuka Public Palaeoanthropology Project, 2026).
Will any building or structure older than 60 years be affected in any way?	NO
Is it necessary to apply for a permit in terms of the National Heritage Resources Act, 1999 (Act 25 of 1999)?	NO
If yes, please submit or, make sure that the applicant or a specialist submits the necessary application to SAHRA or the relevant provincial heritage agency and attach proof thereof to this application if such application has been made.	

SECTION C: PUBLIC PARTICIPATION

1. ADVERTISEMENT

The person conducting a public participation process must take into account any guidelines applicable to public participation as contemplated in section 24J of the Act and must give notice to all potential interested and affected parties of the application which is subjected to public participation by—

- (a) fixing a notice board (of a size at least 60cm by 42cm; and must display the required information in lettering and in a format as may be determined by the competent authority) at a place conspicuous to the public at the boundary or on the fence of—
 - (i) the site where the activity to which the application relates is or is to be undertaken; and
 - (ii) any alternative site mentioned in the application;
- (b) giving written notice to—
 - (i) the owner or person in control of that land if the applicant is not the owner or person in control of the land;
 - (ii) the occupiers of the site where the activity is or is to be undertaken or to any alternative site where the activity is to be undertaken;
 - (iii) owners and occupiers of land adjacent to the site where the activity is or is to be undertaken or to any alternative site where the activity is to be undertaken;
 - (iv) the municipal councillor of the ward in which the site or alternative site is situated and any organisation of ratepayers that represent the community in the area;
 - (v) the municipality which has jurisdiction in the area;
 - (vi) any organ of state having jurisdiction in respect of any aspect of the activity; and
 - (vii) any other party as required by the competent authority;
- (c) placing an advertisement in—
 - (i) one local newspaper; or
 - (ii) any official *Gazette* that is published specifically for the purpose of providing public notice of applications or other submissions made in terms of these Regulations;
- (d) placing an advertisement in at least one provincial newspaper or national newspaper, if the activity has or may have an impact that extends beyond the boundaries of the metropolitan or local municipality in which it is or will be undertaken: Provided that this paragraph need not be complied with if an advertisement has been placed in an official *Gazette* referred to in subregulation 54(c)(ii); and
- (e) using reasonable alternative methods, as agreed to by the competent authority, in those instances where a person is desiring of but unable to participate in the process due to—
 - (i) illiteracy;
 - (ii) disability; or
 - (iii) any other disadvantage.

2. CONTENT OF ADVERTISEMENTS AND NOTICES

A notice board, advertisement or notices must:

- (a) indicate the details of the application which is subjected to public participation; and
- (b) state—
 - (i) that the application has been submitted to the competent authority in terms of these Regulations, as the case may be;

- (ii) whether basic assessment or scoping procedures are being applied to the application, in the case of an application for environmental authorisation;
- (iii) the nature and location of the activity to which the application relates;
- (iv) where further information on the application or activity can be obtained; and
- (iv) the manner in which and the person to whom representations in respect of the application may be made.

3. PLACEMENT OF ADVERTISEMENTS AND NOTICES

Where the proposed activity may have impacts that extend beyond the municipal area where it is located, a notice must be placed in at least one provincial newspaper or national newspaper, indicating that an application will be submitted to the competent authority in terms of these regulations, the nature and location of the activity, where further information on the proposed activity can be obtained and the manner in which representations in respect of the application can be made, unless a notice has been placed in any *Gazette* that is published specifically for the purpose of providing notice to the public of applications made in terms of the EIA regulations.

Advertisements and notices must make provision for all alternatives.

4. DETERMINATION OF APPROPRIATE MEASURES

The practitioner must ensure that the public participation is adequate and must determine whether a public meeting or any other additional measure is appropriate or not based on the particular nature of each case. Special attention should be given to the involvement of local community structures such as Ward Committees, ratepayers associations and traditional authorities where appropriate. Please note that public concerns that emerge at a later stage that should have been addressed may cause the competent authority to withdraw any authorisation it may have issued if it becomes apparent that the public participation process was inadequate.

5. COMMENTS AND RESPONSE REPORT

The practitioner must record all comments and respond to each comment of the public before the application is submitted. The comments and responses must be captured in a comments and response report as prescribed in the EIA regulations and be attached to this application. The comments and response report must be attached under Appendix E.

6. AUTHORITY PARTICIPATION

Authorities are key interested and affected parties in each application and no decision on any application will be made before the relevant local authority is provided with the opportunity to give input. The planning and the environmental sections of the local authority must be informed of the application at least 30 (thirty) calendar days before the submission of the application.

List of authorities informed:

- Eastern Cape Department of Economic Development, Environmental Affairs and Tourism (DEDEAT)
- Department of Water and Sanitation (DWS)
- Eastern Cape Provincial Heritage Resources Authority (ECPHRA)
- Eastern Cape Parks and Tourism Agency (ECPTA)
- South African National Roads Agency (SANRAL)
- Department of Transport (DoT)
- Mbhashe Local Municipality
- Amathole District Municipality

Also refer to the attached I&APs database.

List of authorities from whom comments have been received:

No comments have been received to date.

7. CONSULTATION WITH OTHER STAKEHOLDERS

Note that, for linear activities, or where deviation from the public participation requirements may be appropriate, the person conducting the public participation process may deviate from the requirements of that subregulation to the extent and in the manner as may be agreed to by the competent authority.

Any stakeholder that has a direct interest in the site or property, such as servitude holders and service providers, should be informed of the application at least 30 (thirty) calendar days before the submission of the application and be provided with the opportunity to comment.

Has any comment been received from stakeholders?

☒ YES	☐ NO
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If "YES", briefly describe the feedback below (also attach copies of any correspondence to and from the stakeholders to this application):

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SECTION D: IMPACT ASSESSMENT

The assessment of impacts must adhere to the minimum requirements in the EIA Regulations, 2014 as amended, and should take applicable official guidelines into account. The issues raised by interested and affected parties should also be addressed in the assessment of impacts.

1. ISSUES RAISED BY INTERESTED AND AFFECTED PARTIES

List the main issues raised by interested and affected parties.

No comments have been received to date.

Response from the practitioner to the issues raised by the interested and affected parties (A full response must be given in the Comments and Response Report that must be attached to this report):

No comments have been received to date.

2. IMPACTS THAT MAY RESULT FROM THE PLANNING AND DESIGN, CONSTRUCTION, OPERATIONAL, DECOMMISSIONING AND CLOSURE PHASES AS WELL AS PROPOSED MANAGEMENT OF IDENTIFIED IMPACTS AND PROPOSED MITIGATION MEASURES

List the potential direct, indirect and cumulative property/activity/design/technology/operational alternative related impacts (as appropriate) that are likely to occur as a result of the planning and design phase, construction phase, operational phase, decommissioning and closure phase, including impacts relating to the choice of site/activity/technology alternatives as well as the mitigation measures that may eliminate or reduce the potential impacts listed.

Alternative (preferred alternative)

The potential impacts discussed below have been rated using a standard impact rating method. This method is described in Appendix H. The complete rating table is also included in Appendix H.

IMPACT ASSESSMENT METHODOLOGY

The impacts identified were assessed according to the criteria outlined below. Each impact was ranked according to the nature, extent, duration, magnitude, probability, irreplaceable loss of resources and reversibility. These criteria are based on the Department of Economic Development, Environmental Affairs and Tourism (DEDEAT) Guideline Document to the EIA Regulations (1998). A significance rating of each identified impact was further reviewed by the EAP and/or specialist by applying professional judgement. Where possible, mitigatory measures were recommended for the impacts identified.

Rating	Definition of Rating	Score
A. Spatial extent – the area in which the impact will be experienced		
None		0
Local	Confined to project or study area or part thereof (e.g. site)	1

Regional	The region, which may be defined in various ways, e.g. cadastral. Catchment, topographic	2
(Inter) national	Nationally or beyond	3
B. Intensity – the magnitude or size of the impact		
None		0
Low	Natural and/or social functions and processes continue albeit in a slightly modified way	1
Medium	Natural and/or social functions and processes continue albeit in a modified way	2
High	Natural and/or social functions and processes are severely altered	3
C. Duration - the time frame for which the impact will be experienced		
None		0
Short-term	Up to 2 years	1
Medium-term	2 to 20 years	2
Long-term	More than 20 years	3

Combined Score (A+B+C)	
Consequence Rating	
Not significant	0 - 2
Very low	3 – 4
Low	5
Medium	6
High	7
Very high	8 - 9

Probability – the likelihood of the impact occurring		
Improbable	< 40% chance of occurring	0
Possible	40% - 70% chance of occurring	1
Probable	> 70% - 90% chance of occurring	2
Definite	> 90% chance of occurring	3

Significance Rating	Possible Impact Combinations	
	Consequence	Probability
Insignificant	Very Low	Improbable
	Very Low	Possible
Very Low	Very Low	Probable
	Very Low	Definite
	Low	Improbable
	Low	Possible
Low	Low	Probable
	Low	Definite
	Medium	Improbable
	Medium	Possible
Medium	Medium	Probable

	Medium	Definite
	High	Improbable
	High	Possible
High	High	Probable
	High	Definite
	Very High	Improbable
Very High	Very High	Possible
	Very High	Probable
	Very high	Definite

Status of impact

The environmental impacts of an activity are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the activity not been undertaken. It is an appraisal of the type of effect the activity would have on the affected environmental parameter. Its description includes what is being affected, and how

Indication whether the impact is adverse (negative) or beneficial (positive).	+ ve (positive – a 'benefit')
	– ve (negative – a 'cost')

Confidence of assessment

The degree of confidence in predictions based on available information, EAP's judgement and/or specialist knowledge	Low
	Medium
	High

IMPACTS

PLANNING AND DESIGN PRE-CONSTRUCTION PHASE

Table 1. Impacts identified for the planning and pre-construction phase.

Impacts	Description
Direct	
Poor site establishment	Inappropriate siting of construction camps, stockpiles and laydown areas (e.g. within watercourse buffers or sensitive habitats) can result in unnecessary habitat destruction, erosion, water contamination and inefficient construction activities
Legal and social compliance	Commencement of project construction and activities without lodging and obtaining relevant authorisation applications and being approved for resulting in authorities halting the project, issuing fines, implying no development commencement.
Damage to existing civil services, infrastructure and properties	Disruption and damage of water supply, network and internet, infrastructure and other properties in close proximity to the proposed site.
Environmental awareness training	Lack of environmental awareness, education, on contractor staff and subsequent environmental incidents and or infringements
Traffic congestion and flow	Inadequate planning for access, parking and lack of a traffic management plan may lead to congestion.
Inadequate stormwater management design	Poor stormwater planning may cause erosion, flooding and sedimentation of surrounding areas once construction and operation commence.
Bridge layout hydrological and hydraulic impacts	Lack of a hydraulic data on a bridge layout design can lead to the construction of a bridge that is susceptible to the erosive power of water during flooding This can perpetuate the collapse of the bridges which can yield fatal accidents and structural damage.

Construction Programme	The absence of a construction programme can result in delays on the project. Unexpected issues may occur during construction courtesy of inefficient planning which may result in pronounced delays and possible project termination.
Indirect	
Safety of surrounding community	Lack of incorporation of safety considerations into the construction site layout may induce significant safety risks to road users and surrounding communities. Failure to include safety measures at design stage to sustainably protect municipal infrastructure from theft and vandalism.
Delayed project implementation and social unrests	Inappropriate planning of the tasks and responsible persons may lead to the delayed implementation of the project.
Cumulative	
Sewage overflow pollution	The conference centre introduces additional load to the deteriorating sewer infrastructure of the municipality and may lead into further sewer related pollution into the receiving environment

CONSTRUCTION PHASE

Table 2. Impacts identified for the construction phase.

Impacts	Description
Direct	
Legal and environmental non-compliance	Failure to comply with environmental authorisations, EMP requirements and construction regulations may result in penalties, halting of construction, and overall disapproval and failure of the project.
Site establishment and security	The establishment of a construction site requires appropriate planning to achieve a holistic balance between efficacy and cost-efficiency of construction activities as well as minimised environmental damage.
Damage to existing civil services, infrastructure and properties	Excavations and construction activities may damage water, sewer, electricity and telecommunications infrastructure, causing severe interruptions.
Environmental awareness and training	Lack of environmental awareness and environmental non-compliance of the staff and foremen will result in the damage to environmentally sensitive areas and possible environmental risks associated with construction activities.
Poor waste management	Inadequate handling of construction waste may result in littering and pollution of the immediate and surrounding receiving environment.
Vegetation clearing and habitat loss	Uncontrolled vegetation clearance may lead to the loss of natural vegetation and habitats.
Loss of ecosystem function and connectivity	Loss or alteration of natural habitats, thereby reducing the ability of ecosystems to maintain their ecological functions.
Material stockpiling and dust pollution	Construction activities such as earthworks and construction vehicles might increase the dust particles in the area surrounding the construction site
Hazardous material storage and handling	Improper storage or handling of hazardous materials may pose risks to health and the environment.
Noise pollution	The noise levels in the area may be increased because of the construction activities. Increased traffic from local motorists and construction vehicle might also increase the noise pollution in the receiving environment.
Stormwater management	Failure to implement effective stormwater management measures may result in an increase in surface soil erosion and sedimentation of the nearby watercourse and wetland.

Contamination of surface and groundwater from hazardous substances	Pollution of sediment from sewer, fuel, oil, cement or chemical spillage. Organic pollution of soils and ecosystems resulting in a loss of permeability of soils.
Proliferation of invasive alien vegetation	Uncontrolled vegetation clearance during the fence construction may lead to loss of natural vegetation and habitats. Opportunistic alien and invasive vegetation encroachment may result in post degradation of native vegetation.
Water consumption	Failure to consider technologies that reduce consumption of water can place further pressure on already stressed water resources. Construction activities may place pressure on local water supplies.
Energy consumption and emissions	Fuel consumption by machinery contributes to air pollution and greenhouse gas emissions.
Loss/disturbance of riparian vegetation and associated habitat	Clearing and disturbance of riparian vegetation during construction may result in the localised loss of vegetation cover and associated habitat, with potential effects on bank stability and habitat structure.
Disturbance and modification of the riverbed and banks	Earthworks, excavation, machinery movement and construction of the proposed crossing and associated infrastructure may physically disturb the riverbed and banks and result in localised alteration of the natural channel form and habitat characteristics.
Bank erosion and sedimentation	Vegetation clearing, exposed soils, earthworks and concentrated runoff may increase erosion of the riverbanks and surrounding disturbed areas, resulting in increased sediment input and deposition within the Mputi River and potentially within the nearby artificial dam/impoundment through sediment-laden runoff.
Flow modification	Construction of the proposed crossing may alter the natural movement and distribution of surface flow within the Mputi River through temporary obstruction, diversion or concentration of flow. If not appropriately designed and constructed, the crossing may also cause localised scour or sediment deposition and affect longitudinal ecological connectivity within the river.
Deterioration of surface water quality	Construction activities may introduce sediment, hydrocarbons, chemicals, litter or other pollutants into the river through accidental spills, runoff or inappropriate waste handling, potentially affecting aquatic habitat.
Palaeontological and heritage resource disturbance	Damage to cultural, heritage, archaeological and paleaontological resources during construction activities such as excavations and plant movement.
Traffic disruption	Temporary disturbance for pedestrian and livestock traffic. Normal traffic flow in the area will be disturbed. This might lead to slower traffic flow resulting from increased construction vehicles in the area. This might also lead to increased pedestrian, animal and vehicle accidents.
Employment and business opportunities	Temporary employment opportunities will be created for the local's inhabitants and SMME's. In the process, the beneficiaries will accrue source of income and be upskilled in construction methods. Thereby, improving livelihoods of the inhabitants.
Cumulative	
Vegetation clearing, excavations and habitat loss	Uncontrolled vegetation clearance and horizontal alignment encroachment into undisturbed areas may lead to loss of natural vegetation and habitats. Opportunistic alien and invasive vegetation encroachment may result in post degradation of native vegetation.
Increased downstream overflow and enhanced hillslope and bank channel erosion	Diverting construction water to the neighbouring ephemeral stream might impede or divert the flow of water in the stream therefore altering the bed, banks, courses or characteristics of the watercourse. Perpetual soil erosion, bank erosion, channel scouring, and in stream sedimentation may be induced by construction activities.
Natural vegetation resprouting	Failure and inadequate re-vegetation of disturbed areas where construction is completed may lead to environmental degradation of the affected environment.
Invasive alien vegetation	Uncontrolled vegetation clearance may lead to loss of natural vegetation and habitats. Opportunistic alien and invasive vegetation encroachment may result in post degradation of native vegetation.

Disturbance on aquatic biodiversity	The proposed development may contribute cumulatively to existing pressures on the Mputi River, including road crossings, stormwater inputs, bank erosion and solid-waste accumulation. Additional habitat disturbance, runoff, sedimentation and pollution associated with the development may therefore contribute to progressive degradation of instream and riparian habitat.
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REHABILITATION PHASE

Table 3. Impacts identified for the rehabilitation phase.

Impacts	Description
Direct	
Inadequate environmental compliance	Lack of post construction follow-up on environmental obligations fulfilment may result into unattended environmental impacts which may lead to environmental degradation.
Poor site de-establishment	Environmental pollution such as illegally dumped waste and construction material might be a result of unmonitored site de-establishment.
Inadequate re-vegetation of disturbed areas	Failure to re-establish vegetation may leave soils exposed, increasing erosion and reducing ecological recovery.
Visual and landscape degradation	Poor rehabilitation may negatively affect the aesthetic quality of the conference centre environment and surrounding area.
Indirect	
Downslope and downstream stormwater-sedimentation channel	Lack of implementation of appropriate soil erosion and slope stabilisation methods may lead to alarming rates of soil erosion on the excavated areas, and other that are not vegetated.
Increased maintenance burden	Inadequate rehabilitation may increase long-term maintenance costs for the conference centre and the municipality.
Cumulative	
Long-term degradation of natural vegetation	Failure and inadequate re-vegetation of disturbed areas where construction is completed may lead to environmental degradation of indigenous vegetation cover of the affected environment.
Increased prevalence invasive alien vegetation	Uncontrolled vegetation clearance may lead to loss of natural vegetation and habitats. Opportunistic alien and invasive vegetation encroachment may result in post degradation of native vegetation.

OPERATIONAL PHASE

Table 4. Impacts identified for the operational phase.

Impacts	Description
Direct	
Legal and environmental non-compliance	Failure to fulfil legal environmental legislations and EA conditions during operations may lead to overall disapproval and failure of the project
Safety and security	Safety measures are needed to ensure the safety of the staff, learners and school property.
Social and economic benefits	There may be increased permanent employment opportunities, skills development and an increase in which may stimulate the local economy.
Sanitation and hygiene	Poorly maintained ablution facilities/sewerage pipelines may result in the pollution of the immediate and surrounding environment and poor maintenance may cause health risks
Freshwater consumption	Ongoing water use for sanitation, cleaning and landscaping may place pressure on municipal water resources.

Solid waste generation	Daily school activities generate waste which may pollute the environment if not managed properly. The lack of waste handling and management may result into littering, illegal dumping and pollution of the immediate and surrounding receiving environment.
Energy consumption	Inappropriate use of energy could lead to a reduction of this source and increased operational costs if inefficient.
Greening	Vegetation and disturbed soil rehabilitation will contribute to soil organic carbon and CO ₂ re-sequestration.
Stormwater management	Ineffective stormwater systems may result in soil erosion and localised flooding.
Increased stormwater-runoff	Permanent hard surfaces may increase the volume and concentration of stormwater runoff during rainfall events, increasing erosion and the potential for sediment-laden runoff to enter the Mputi River.
Risk of sewer pipeline leakage	Leakage, rupture or failure of the sewer pipeline could result in sewage entering the Mputi River, potentially causing deterioration of water quality and associated impacts on aquatic and riparian habitat.
Ongoing disturbance of riparian habitat	Routine maintenance of the infrastructure may result in continued localised disturbance of riparian vegetation and associated habitat.

No-Go

Table 5. Impacts identified for the Status quo.

Impacts	Description
Status quo – No conference centre	Without the proposed development, the lack of a dedicated conference centre and events facility in Dutywa would remain, limiting the municipality's ability to host conferences, training programmes, government meetings, exhibitions and community events within the local area.
Socio-economic	Without the proposed development, temporary employment opportunities during construction and permanent jobs during operation would not be created, while local contractors, suppliers, and other SMMEs would not benefit from the increased economic activity generated by the conference centre. The opportunity to attract business tourism, investment and visitor spending to Dutywa would also be lost.
Local development	The absence of a modern conference centre and events facility would limit opportunities for government service delivery, business networking, community engagement and regional investment, thereby constraining the long-term social and economic development of Dutywa and the surrounding communities. The opportunity to improve local infrastructure would be forgone.

SUMMARY OF SPECIALIST STUDIES IDENTIFIED IMPACTS

Specialist studies are appended as **Appendix D** of this report. This section provides an overview of the key findings from the conducted specialist studies (excluding the Cultural and Heritage outlined in Section 6 of this report);

- Aquatic Biodiversity Impact Assessment (Abantu Environmental Services, 2026)
- Terrestrial Biodiversity Impact Assessment (Mbambo Ecological Services, 2026)
- Palaeontological Impact Assessment (Umsuka Public Palaeoanthropology Project, 2026)
- Archaeological and Cultural Heritage Impact Assessment (Umsuka Public Palaeoanthropology Project, 2026)

AQUATIC BIODIVERSITY ASSESSMET

Key Findings

- The proposed development site is located within aquatic ESA 1.
- According to the NFEPA datasets the development does not overlap any FEPA rivers. The NFEPA dataset identifies a mapped wetland feature within 500 m of the proposed development footprint. Field verification confirmed that the mapped aquatic feature within this area comprises an artificial dam/impoundment rather than a natural wetland system.
- The National Wetland Map version 5 (NWM5) shows no inland wetland ecosystems.
- The proposed development footprint does not fall within any SWSA.
- The Index of Habitat Integrity (IHI) assessment for the Mputi River indicates that the instream component falls within Integrity Class C (Moderately Modified), while the riparian component falls within Integrity Class B (Largely Natural with Few Modifications).
- The ecological importance and sensitivity category of the Mputi River reach was determined as being Moderate.

Mitigation and Recommendations

- Obtain all required environmental and water-related approvals prior to any site works.
- Limit vegetation clearing to the area required for construction.
- Clearly demarcate the approved working footprint and treat areas outside it as no-go areas.
- Locate construction camps, laydown areas, and materials stockpiles outside of the riparian zone.
- Use existing access routes where practicable.
- Avoid unnecessary removal of indigenous riparian vegetation.
- Rehabilitate disturbed areas immediately after construction.
- Excavation should be limited only to the approved footprint.
- Restore disturbed banks after construction and stabilise exposed areas to prevent further disturbance.
- Implement erosion and sediment-control measures.
- Stabilise exposed soils and locate stockpiles away from the Mputi River, artificial dam/impoundment and associated drainage pathways.
- Works within the active channel should, where practicable, be undertaken during naturally low-flow conditions.
- Disturbance within the channel must be restricted to the development footprint.
- Temporary flow diversions, where required, must be removed after construction.
- Minimise works within the watercourses as far as practically possible.
- Refueling and maintenance of machinery must take place away from the watercourse.
- Mixed cement must not be allowed to flow into any watercourses.
- Drip trays must be placed under stationary construction machinery overnight to avoid soil contamination from oil and fuel leaks
- Spill-response kits must be available on site.
- Restrict all works to the construction footprint.
- Where possible, existing disturbed areas should be used for access and working areas.
- Rehabilitate disturbed areas immediately after construction.
- Prevent direct discharge of concentrated runoff into the Mputi River.
- Stormwater infrastructure must be regularly inspected and maintained.
- Sewer pipeline must be designed and installed to minimise the risk of leakage at the river crossing.
- Regular inspection and maintenance must be undertaken.
- Any leak or failure must be repaired immediately.
- Avoid unnecessary clearing of riparian vegetation.

- Disturbed areas must be rehabilitated after maintenance activities, and alien invasive plants should be controlled where they establish.

TERRESTRIAL BIODIVERSITY IMPACT ASSESSMENT

Key Findings

- The proposed development site is located in Mthatha Moist Grassland which is listed as an Endangered ecosystem type in the List of Threatened Ecosystems (2021) and the SANBI's Red list of Ecosystems.
- The proposed development site does not occur within or near any protected areas or its formal buffer zone; therefore, direct physical impacts are not expected.
- The proposed development falls within an upstream management area.
- There are no forest patches within the development footprint.
- No plant Species of Conservation (SCC) were identified on site and by the Screening Tool. In addition, the NEWPOSA database was consulted, which indicated that no SCCs are expected to occur within the project area. All the plant species identified on site are listed as Least Concern.
- No extensive bare areas or visible soil erosion were observed within the conference-centre footprint.

Mitigation and Recommendations

- Minimise natural vegetation clearance: Limit the footprint of disturbance by maximising the use of existing tracks, roads, and infrastructure corridors wherever possible.
- Environmental supervision: Appoint a competent Environmental Control Officer (ECO) for the construction phase to ensure all recommendations in this report are implemented.
- Restrict access: Limit all construction and maintenance access to demarcated areas to prevent disturbance to vegetation outside the development footprint.
- Protect natural resources: Prohibit collection of plants, wood, or other natural resources outside the approved development footprint.
- Laydown areas: Restrict laydown, storage, and temporary work areas to sites of low ecological sensitivity to reduce unnecessary habitat loss.
- Selective clearing: Only clear vegetation that is essential for construction, such as for pipelines and avoid unnecessary widening of access routes, laydown areas, material storage areas and construction camps.
- Topsoil management: Remove and stockpile topsoil carefully during construction for use in the rehabilitation of disturbed areas.
- Rehabilitation timing: Undertake rehabilitation immediately after construction activities cease in a given area to reduce erosion and promote ecosystem recovery.
- Rehabilitation methods: Allow mostly passive regeneration of vegetation, supplemented initially with indigenous grass mixes to stabilise soils and accelerate recovery.
- Protected species management: Apply for relevant permits prior to removing any protected species, and ensure removal is conducted in accordance with legal requirements.
- Secondary and Tertiary PAOI are no-go areas, and no development activities are allowed.
- Monitor alien invasive plant species during construction and implement routine removal of the species
- Where required, rehabilitate and revegetate areas as soon as possible using indigenous plant species
- Implement the Open Space Maintenance Plan
- Limit the use of pesticides and herbicides during the operation phase, particularly with the control of AIS
- Access should be limited to prevent further disturbance to the vegetation outside of the development footprint
- No fires should be allowed on site

- No plant or wood collection should be allowed outside of the proposed development footprint
- Prevent development from occurring within the identified No Go Areas (i.e surrounding vegetation)

PALAEONTOLOGICAL IMPACT ASSESSMENT

Key findings

- The proposed Mbashe Conference Centre and associated infrastructure are located on sedimentary rocks of the Katberg Formation, Tarkastad Subgroup, Beaufort Group, Karoo Supergroup.
- The formation possesses significant regional palaeontological potential. However, fossils are not uniformly distributed throughout the succession.
- No known fossil locality has been identified within the development footprint on the basis of the information considered during this desktop assessment. Nevertheless, the possibility of encountering isolated palaeontological resources during construction cannot be excluded.

Mitigation and Recommendations

- A Fossil Chance Find Protocol should nevertheless be incorporated into the development as a precaution. If suspected fossil material is exposed, work must stop locally, the find must be protected and a qualified palaeontologist or SAHRA must be notified before work resumes in the affected area.
- The Environmental Compliance Officer responsible for overseeing the project should be informed of the geological context of the site and the extremely low likelihood of encountering palaeontological material.
- Construction managers and supervisory personnel should be made aware of the procedures to follow in the unlikely event that fossil material is encountered.
- Any new geological exposures created during excavation for foundations, cable trenches, substations, or access roads should be visually inspected as part of routine environmental monitoring.
- In the unlikely event that fossil material is discovered, work in the immediate vicinity must cease and a qualified palaeontologist or the relevant heritage authority must be contacted for further guidance.
- Any palaeontological material identified as scientifically significant must be recorded and sampled by a qualified specialist. The costs associated with this activity shall be borne by the developer as part of responsible heritage management.

ARCHAEOLOGICAL AND CULTURAL HERITAGE IMPACT ASSESSMENT

Key findings

- There is no archaeological record of surveys or excavations from the location proposed for development, based on the records from SAHRIS, and academic publications.
- There is no evidence of archaeological excavations, surveys, or mitigation from the development area.
- The area is rated as 'General' Protection C (Field Rating IV C) and does not need to be mitigated.
- There is also no evident archaeological or heritage remains found in the development area based on the satellite images and maps.
- The area is considered as having a low archaeological heritage significance.

Mitigation and Recommendations

- If any archaeological material and/or human remains (including graves and burials) are uncovered during construction, all work must cease immediately. These finds must be reported to the Albany Museum (046 622 2312) and/or the Eastern Cape Provincial Heritage Resources Agency (ECPHRA) (043 745 0888) so that systematic and professional investigation/excavation can be undertaken.

- Construction managers/foremen and/or the Environmental Control Officer (ECO) should be informed before construction starts on the possible types of heritage sites and cultural material they may encounter and the procedures to follow when they find sites.
- A Palaeontological Impact Assessment (PIA) is being conducted alongside this AIA and any recommendations there also need to be considered and followed.

IMPACT RATINGS

A summary of all the identified impacts, their associated phase, as well as their impact calculations and significance are presented in (**Table 6**).

Table 6.Impact ratings for each development phase.

	Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
	Direct											
Planning and Design	Poor site establishment	Without	1	3	1	5	Low	3	Definite	Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Legal and social compliance	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Damage to existing civil services, infrastructure and properties	Without	1	3	2	6	Medium	2	Probable	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Environmental awareness training	Without	1	2	2	5	Low	2	Probable	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Traffic congestion and flow	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Inadequate stormwater management design	Without	1	3	2	6	Medium	3	Definite	Medium	-ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	-ve	High
	Bridge layout hydrological and hydraulic impacts	Without	1	2	2	5	Low	3	Definite	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Construction Programme	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Indirect Impacts											
	Safety of surrounding community	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Delayed project implementation and social unrests	Without	2	3	2	7	High	2	Probable	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Cumulative Impacts											
	Sewage overflow pollution	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
Construction	Direct											
	Legal and environmental non-compliance	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Site establishment and security	Without	1	3	2	6	Medium	2	Probable	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Damage to existing civil services, infrastructure and properties	Without	1	3	2	6	Medium	2	Probable	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Environmental awareness and training	Without	1	3	2	6	Medium	2	Probable	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Poor waste management	Without	1	3	2	6	Medium	2	Probable	Medium	- ve	High

	Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Vegetation clearing and habitat loss	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Loss of ecosystem function and connectivity	Without	1	3	2	6	Medium	3	Definite	Medium	-ve	High
		With	1	2	1	4	Very low	2	Probable	Very Low	-ve	High
	Material stockpiling and dust pollution	Without	1	1	1	3	Very low	3	Definite	Very Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Hazardous material storage and handling	Without	1	2	2	5	Low	3	Definite	Low	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Noise pollution	Without	1	2	2	5	Low	3	Definite	Low	+ ve	High
		With	1	1	2	4	Very low	1	Possible	Insignificant	- ve	High
	Stormwater management	Without	1	3	1	5	Low	3	Definite	Low	+ ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Contamination of surface and groundwater from hazardous substances	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	1	2	1	4	Very low	2	Probable	Very Low	- ve	High
	Proliferation of invasive alien vegetation	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	1	2	1	4	Very low	2	Probable	Very Low	- ve	High
	Water consumption	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Energy consumption and emissions	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Loss/disturbance of riparian vegetation and associated habitat	Without	1	2	2	5	Low	2	Probable	Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Disturbance and modification of the riverbed and banks	Without	1	2	2	5	Low	3	Definite	Low	- ve	High
		With	1	2	1	4	Very low	1	Possible	Insignificant	- ve	High
	Bank erosion and sedimentation	Without	1	2	2	5	Low	3	Definite	Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Flow modification	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Deterioration of surface water quality	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Palaeontological and heritage resource disturbance	Without	1	3	3	7	High	2	Probable	High	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Traffic disruption	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	2	1	4	Very low	2	Probable	Very Low	- ve	High
	Employment and business opportunities	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
		With	2	3	2	7	High	3	Definite	High	+ ve	High

	Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
	Cumulative Impacts											
	Vegetation clearing, excavations and habitat loss	Without	1	3	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Increased downstream overflow and enhanced hillslope and bank channel erosion	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Natural vegetation reporting	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Invasive alien vegetation	Without	1	3	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Disturbance on aquatic biodiversity	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
Rehabilitation	Direct											
	Inadequate environmental compliance	Without	2	2	2	6	Medium	2	Probable	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Poor site de-establishment	Without	1	3	3	7	High	2	Probable	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Inadequate re-vegetation of disturbed areas	Without	1	3	3	7	High	2	Probable	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Visual and landscape degradation	Without	1	3	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Indirect Impacts											
	Downslope and downstream stormwater- channel sedimentation	Without	2	3	2	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Increased maintenance burden	Without	2	2	3	7	High	1	Possible	Medium	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Cumulative Impacts											
	Long-term degradation of natural vegetation	Without	1	3	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Increased prevalence invasive alien vegetation	Without	1	3	2	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
Operations	Direct											
	Legal and environmental non-compliance	Without	1	2	2	5	Low	2	Probable	Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Safety and security	Without	1	2	2	5	Low	3	Definite	Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	- ve	High
	Social and economic benefits	Without	1	2	3	6	Medium	2	Probable	Medium	- ve	High

	Impact	Mitigation	Extent	Intensity	Duration	Consequence	Consequence	Probability	Probability	Significance	Status	Confidence
		With	1	3	3	7	High	3	Definite	High	+ve	High
	Sanitation and hygiene	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Freshwater consumption	Without	2	2	3	7	High	3	Definite	High	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Solid waste generation	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Energy consumption	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	1	1	3	Very low	2	Probable	Very Low	- ve	High
	Greening	Without	1	2	3	6	Medium	3	Definite	Medium	- ve	High
		With	1	3	3	7	High	3	Definite	High	+ ve	High
	Stormwater management	Without	1	2	1	4	Very low	3	Definite	Very Low	- ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Increase in anthropogenic disturbance	Without	1	2	2	5	Low	2	Probable	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Increased stormwater runoff	Without	1	2	2	5	Low	2	Probable	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Risk of sewer pipeline leakage	Without	1	2	2	5	Low	2	Probable	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
	Ongoing disturbance of riparian habitat	Without	1	2	2	5	Low	2	Probable	Low	-ve	High
		With	1	1	1	3	Very low	1	Possible	Insignificant	-ve	High
No-Go	Status-quo – No conference centre	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High
	Socio-economic	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High
	Local development	Without	2	2	2	6	Medium	3	Definite	Medium	- ve	High
		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	High

MITIGATIONS

PLANNING AND DESIGN PRE-CONSTRUCTION PHASE

Table 7. Mitigation measures for the planning and pre-construction phase.

Aspects	Mitigation measures
Direct	
Poor site establishment	• Undertake proper site investigations and surveys prior the development. • Clearly define the development footprint and site boundaries before site establishment. • Ensure the site layout is aligned with approved architectural and engineering designs. • Avoid development in environmentally sensitive areas.
Legal and social compliance	• Ensure environmental authorisations namely EA are obtained for the development. • Environmental Control Officer (ECO) to be appointed to monitor compliance with permits conditions. • All health and safety permits and certification to be in place • Appointment of a Community Liaison Officer (CLO) . • Appointment of a Social Facilitator (SF) to manage social-economic aspects of the project relating to community and local business interest, particularly small, medium and micro-enterprise (SMMs). • A health and safety plan in terms of the Occupational Health and Safety Act (Act No 85 of 1993) must be drawn up by the HSE officer prior to construction to ensure workers safety. • Environmental files to be compiled prior to commencement of construction activities. • Methods statement to be compiled for all works with environmental aspects and impacts, for approval of the ECO. • Employment and sub-contracting of locals.
Damage to existing civil services, infrastructure and properties	• Ensure that all existing services and infrastructure are known, mapped and marked to prevent potential damage. • Development of a plan to manage existing services to avoid damages. • Wayleaves to be obtained from the relevant custodians for affected servitudes as applicable. • Property owners such as the local residents should be engaged throughout the project lifespan and engaged as necessary to promote conflict free development implementation. • Where relevant, the directly affected property owners should be informed in writing and engaged to reach a consensus and those proceedings and relevant agreements to be documented.
Environmental awareness and training	• An environmental education and awareness plan must be developed for the project. • Immediately after the site establishment phases, environmental education and awareness are to be conducted for site staff. • The training session shall be conducted, as far as possible, in the employees' language of choice. • As a minimum, training shall include: ○ Explanation of the importance of complying with the EMP, EA and WULA; ○ Discussion of the potential environmental impacts of construction activities; ○ Explanation of the mitigation measures that must be implemented when carrying out their activities; ○ Explanation of the specifics of this EMP and its implementation; and ○ Explanation of the management structure of individuals responsible for matters pertaining to the EMP.

Aspects	Mitigation measures
Traffic congestion and flow	• Development of a Traffic Management Plan that must be implemented throughout the construction and operational phase. • A Traffic Impact Assessment is recommended for safety of the development prior to implementation • Pedestrians and motorists should be accommodated through the construction through appropriate traffic warning signs and measures • Design safe and clearly separated vehicle and pedestrian access routes • Provide dedicated drop-off and pick-up zones. • Provide traffic calming measures such as speed humps and pedestrian crossings.
Bridge layout hydrological and hydraulic impacts	• A well designed, comprehensive hydraulics study needs to be conducted and incorporated on the bridge design to reduce to probability of bridge failure. • Engineers need to factor in hydraulic metrics such as flow velocity, flood levels, sediment accumulation and scour potential in order to design bridges that can withstand these forces. • Design a layout that possess minimal hydrological and hydraulic impacts
Construction Programme	• Construction programmes needs to be designed where there is a clear outline of the project scope, timeline, development phases. • A clear indication of the activities and materials needed for the project must be incorporated in the construction programme. This must include the budgets and cashflow associated with the project activities. • Where possible works must be carried out during no-low flow periods i.e. winter and autumn to early spring.
Energy consumption	• The design should consider the use of alternative renewable energy sources such as solar power. • Design buildings to be energy efficient (natural lighting and ventilation). • Use energy-efficient lighting and equipment.
Inadequate stormwater management design	• Develop a storm water drainage design during detail design development stage. • Storm water management designs and slope stabilisation must be in accordance with generally accepted design standards • A storm water management plans or mitigation to be developed prior to commencement of construction activities
Habitat loss and surface water disturbance	• Buffers around nearby watercourses to be marked off as no-go areas • Prior to commencement of construction, a floral search and rescue operation should be conducted by a suitably qualified SACNASP registered specialist • Develop a rehabilitation plan by a suitably qualified SACNASP registered specialist to foster phased rehabilitation during and post construction
Inadequate rehabilitation and maintenance of the natural environment	• Develop a rehabilitation plan to foster phased rehabilitation during and post construction.
Cumulative	
Sewage overflow pollution	• Confirm capacity of existing sewer infrastructure. • Upgrade connections where necessary. • Ensure proper design and maintenance of sanitation systems. • Implement regular inspection and maintenance programmes.

CONSTRUCTION PHASE

Table 8. Mitigation measures for the construction phase.

Aspects	Mitigation measures
	Direct
Legal and environmental non-compliance	• ECO to be appointed to audit environmental compliance with conditions in the EMP and other permits and authorisations. • CLO and Social Facilitator to be retained throughout the construction tenure to represent the surrounding communities and local business, particularly SMMEs.
Site establishment and security	• Construction site to be selected by the contractor in conjunction with the consulting engineer and the ECO. • Fulltime security officers to be appointed to safeguard the site, its personnel, plant and equipment. • The site to be clearly and firmly demarcated using appropriate methods. • No-go areas to be clearly and firmly barricaded using appropriate techniques. • Necessary health and safety as well as road construction signage to be installed.
Damage to existing civil services, infrastructure and properties	• Ensure that all existing services and infrastructure are known, mapped and marked to prevent potential damage. • Maintain complaints and incident registers on site throughout the project duration.
Environmental awareness and training	• Undergo environmental induction for all staff and foremen prior to commencement of work on site. • A training programme to be presented by ECO on possible environmental risks that may result from construction activities and how to deal with these (including a reporting structure) must be made available prior to construction commencing. • Undergo internal health and safety and environmental training (induction) to cover all identified project impacts and linked activities.
Poor waste management	• Provision of waste bins for general and hazardous waste. • Waste bins should be marked accordingly and have waterproof lids. • Littering and dumping are prohibited. • Development of a waste disposal register to serve as proof of safe disposal of waste generated from site activities. • Promote recycling activities through provision of waste bins for recyclables which may be dominant waste types such as metal, plastics etc, conditionally that the site produces adequate recyclables. • Ablutions to be kept clean with necessary needs and must be regularly serviced by a certified service provider and manifests be kept on site.
Clearing of indigenous vegetation and habitat loss	• Minimise vegetation clearance and confine to the infrastructural footprint. • Implement systematics and incremental rehabilitation. • Regular monitoring of implementation of this plan for the rehabilitation of disturbed areas must be conducted throughout the duration of the construction phase. • Limit the footprint of disturbance by maximizing the use of existing tracks, roads, and infrastructure corridors wherever possible. • Appoint a competent Environmental Control Officer (ECO) for the construction phase to ensure all recommendations in this report are implemented. • Limit all construction and maintenance access to demarcated areas to prevent disturbance to vegetation outside the development footprint. • Prohibit collection of plants, wood, or other natural resources outside the approved development footprint. • Restrict laydown, storage, and temporary work areas to sites of low ecological sensitivity to reduce unnecessary habitat loss.

Aspects	Mitigation measures
	- Only clear vegetation that is essential for construction, such as for pipelines and avoid unnecessary widening of access routes, laydown areas, material storage areas and construction camps. - Remove and stockpile topsoil carefully during construction for use in the rehabilitation of disturbed areas. - Undertake rehabilitation immediately after construction activities cease in a given area to reduce erosion and promote ecosystem recovery. - Allow mostly passive regeneration of vegetation, supplemented initially with indigenous grass mixes to stabilise soils and accelerate recovery. - Apply for relevant permits prior to removing any protected species, and ensure removal is conducted in accordance with legal requirements.
Loss of ecosystem function and connectivity	- Minimise natural vegetation clearance as far as possible by minimising the development footprint; avoiding unnecessary clearing or disturbance of surrounding grassland; and avoiding unnecessary widening of the road. - Secondary and Tertiary PAOI are no-go areas, and no development activities are allowed.
Material stockpiling and dust pollution	- Stockpiles of excavated earth must not exceed 2 m height and should be covered to avoid wind induced erosion. - Implement dust suppression measures as required. - Vehicle speed should be kept at a minimal reduce dust pollution - Excavations and other clearing activities must only take place during agreed working times and permitted weather conditions to avoid drifting of sand and dust into neighbouring areas.
Hazardous material storage and handling	- Material storage and handling and hazardous material storage area to comply with all relevant environmental and health and safety requirements. - Methods Safety Datasheets for all hazardous material to be kept on site. - Fines haulage and delivery plants to be covered to prevent dust pollution. - Cement and concrete must only be mixed in designated areas and on an impermeable surface.
Noise pollution	- Ensure that construction vehicles and working machinery are serviced and are in good condition to reduce their noise levels - Construction activities should be maintained during the normal working hours (08h00-17h00) - Pedestrians and motorists should be accommodated through appropriate traffic warning signs and measures - Where works are to be carried outside the normal working hours, the affected adjacent residents must be informed and such be undertaken within the shortest time possible
Stormwater management	- Natural (storm water) runoff must be diverted away from the construction area - The construction site must be managed with silt trap or erosion berms in order to prevent pollution on surface water or groundwater, due to suspended solids, silt or chemical pollutants.
Contamination of surface and groundwater from hazardous substances	- Construction plant should be well serviced and in good condition. - Drip trays should be provided for stationary and over-night parked plant. - Concrete batching should be done on impermeable surfaces not directly onto soil substrate. - Hazardous material spillages must be cleaned-up immediately with the contaminated substrate and be disposed of to a registered landfill site.
Proliferation of invasive alien vegetation	Regular and ad-hoc alien vegetation removal.

Aspects	Mitigation measures
	Construction site maintenance to include vegetation control and removal of alien invasive vegetation from the land and other surrounding areas.
Water consumption	- Water for all construction activities should be sourced from a bona fide source. - Access to clean drinking water should be provided to employees and visitors. - If water is stored on site, drinking water and multi-purpose water storage facilities shall be clearly distinguished and designated. - Water conservation measures must be adopted into the building design. - Rainwater tanks should be considered.
Energy consumption and emissions	During the construction phase the contractors are advised to consider the use of generators.
Loss/disturbance of riparian vegetation and associated habitat	- Limit vegetation clearing to the area required for construction. - Clearly demarcate the approved working footprint and treat areas outside it as no-go areas. - Locate construction camps, laydown areas, and materials stockpiles outside of the riparian zone. - Use existing access routes where practicable. - Avoid unnecessary removal of indigenous riparian vegetation. - Rehabilitate disturbed areas immediately after construction.
Disturbance and modification of the riverbed and banks	- Excavation should be limited only to the approved footprint. - Restore disturbed banks after construction and stabilise exposed areas to prevent further disturbance.
Bank erosion and sedimentation	- Implement erosion and sediment-control measures. - Stabilise exposed soils and locate stockpiles away from the Mputi River, artificial dam/impoundment and associated drainage pathways.
Flow modification	- Works within the active channel should, where practicable, be undertaken during naturally low-flow conditions. - Disturbance within the channel must be restricted to the development footprint. - Temporary flow diversions, where required, must be removed after construction.
Deterioration of surface water quality	- Minimise works within the watercourses as far as practically possible. - Refueling and maintenance of machinery must take place away from the watercourse. - Mixed cement must not be allowed to flow into any watercourses. - Drip trays must be placed under stationary construction machinery overnight to avoid soil contamination from oil and fuel leaks - Spill-response kits must be available on site.
Palaeontological and heritage resource disturbance	- Recommendations of the specialists for cultural, heritage and archaeological impact assessment as well as Palaeontological resource assessment should be implemented fully during the course of the project - The stripping of the expansive soil for firmer aggregates and foundation preparation of the rank flooring and drainage construction must be done with precision and careful observation of any possible fossils - Should fossil remains be discovered during the detailed geological inspection, the ECO responsible for the developments would be alerted immediately. - Such discovery must be protected so that a professional palaeontologist will make appropriate mitigation. - If a new paleontological resource is discovered, it is the jurisdiction of the SAHRA to recover and store such resource. Hence, further exploration must be conducted by a professional palaeontologist. - If any archaeological material and/or human remains (including graves and burials) are uncovered during construction, all work must cease immediately. These finds must be reported to the Albany Museum (046 622 2312) and/or the Eastern Cape Provincial

Aspects	Mitigation measures
	Heritage Resources Agency (ECPHRA) (043 745 0888) so that systematic and professional investigation/excavation can be undertaken. Construction managers/foremen and/or the Environmental Control Officer (ECO) should be informed before construction starts on the possible types of heritage sites and cultural material they may encounter and the procedures to follow when they find sites.
Traffic disruption	- Pedestrians and motorists should be accommodated through appropriate traffic warning signs and measures. - Speed should be kept at a minimum next to the construction site.
Employment and business opportunities	Employment of local labour and contracting of SMMEs.
Cumulative	
Vegetation clearing, excavations and habitat loss	- Minimise vegetation clearance and confine to the infrastructural footprint. - Implement systematic and incremental rehabilitation. - Regular monitoring of implementation of this plan for the rehabilitation of disturbed areas must be conducted throughout the duration of the construction phase. - Prevent development from occurring within the identified No-Go areas (i.e. surrounding vegetation).
Increased downstream overflow and enhanced hillslope and bank channel erosion	- Encroachment of the regulated water areas should be minimised - Stormwater cut-offs /attenuation should be implemented to minimise downslope run-off accumulation - Stabilisation of hill slopes through vegetation of angulating land is recommended. - The measures employed must ensure that little to no alteration of the natural flow of the channel. - Material stockpiles must be located outside the 100m zone from any watercourse, and they must be monitored for erosion and alien vegetation infestation
Natural vegetation resprouting	- Re-vegetation of all disturbed and exposed surface with the same or similar indigenous vegetation. - Vegetation cover of at least 80% should be achieved in the construction site.
Invasive alien vegetation	- Maintenance of the receiving areas should include alien vegetation control and eradication. - Monitor alien invasive plant species during construction and implement routine removal of the species. - Where required, rehabilitate and revegetate areas as soon as possible using indigenous plant species.
Disturbance on aquatic biodiversity	- Restrict all works to the construction footprint. - Where possible, existing disturbed areas should be used for access and working areas. - Rehabilitate disturbed areas immediately after construction.

REHABILITATION PHASE

Table 9. Mitigation measures for the rehabilitation phase.

Aspects	Mitigation measures
Direct	
Inadequate environmental compliance	All rehabilitation activities must comply with the approved EMP and conditions of the Environmental Authorisation.
Poor site de-establishment	All construction camps to be left clean, free of litter and any other foreign objects and material relating to construction activities.
Inadequate re-vegetation of disturbed areas	- All disturbed areas must be re-vegetated as soon as practicable after construction. - Indigenous, locally occurring plant species must be used. - Topsoil must be replaced and protected during re-vegetation. - Irrigation or temporary protection measures must be provided where required to ensure successful establishment.
Visual and landscape degradation	- Implement landscaping and rehabilitation in accordance with an approved landscaping plan. - Shape and contour disturbed areas to blend with surrounding landforms. - Remove temporary structures and restore natural visual character where feasible.
Indirect	
Downslope and downstream stormwater-channel sedimentation	- Appropriate erosion control and slope stabilisation measures should be implemented. - Re-vegetation of disturbed and exposed areas with indigenous vegetation.
Increased maintenance burden	- Rehabilitation designs must prioritise low-maintenance, indigenous vegetation. - A post-construction maintenance plan must be developed and implemented. - Responsibilities for maintenance must be clearly assigned to the relevant authority or school management.
Cumulative	
Long-term degradation of natural vegetation	- Re-vegetation of all disturbed and exposed surface with the same or similar indigenous vegetation. - Vegetation cover of at least 80% should be achieved in the construction site.
Increased prevalence invasive alien vegetation	Maintenance of the receiving areas should include alien vegetation control and eradication.

OPERATIONS PHASE

Table 10. Mitigation measures for the operations phase.

Aspects	Mitigation measures
Direct	
Legal and environmental non-compliance	All relevant legislation and policy must be consulted, and the proponent must ensure that the project is compliant with such legislation and policy. The operational conditions outlined in the EA must be adhered to.
Safety and security	- Implement controlled access points, using security systems or personnel to regulate entry. - Develop an Emergency Response Plan.
Social and economic benefits	- Developer to prioritize local skilled and unskilled labour for the operation. - Local contractor should be prioritized for all procurement. - Prioritise local staff and service providers for day-to-day operations.
Sanitation and hygiene	Maintain sanitation facilities in a good and hygienic condition.

Aspects	Mitigation measures
	- Regular inspection faults, leaks and upgrades requirements. - Any leaks and failures of the sewer system must be fixed immediately
Freshwater consumption	- Water-wise practices should be considered such as use of rainwater for non-portable water requirements. - Leak monitoring, detection repairs be implemented.
Solid waste generation	- Compilation of a waste management plan, amongst other things to consider: - Provision of waste bins for general and hazardous waste - Promote recycling activities through provision of waste bins for recyclables which may be dominant waste types such as metal, plastics etc.
Energy consumption	Consider use of alternative energy sources such as solar panels placed on the infrastructure roofs to power the facility.
Greening	Maintain acceptable levels of greenness to offset certain climate impacts.
Stormwater management	Stormwater management infrastructure must be properly maintained and monitored.
Increase in Anthropogenic disturbance	- Implement the Open Space Maintenance Plan. - Limit the use of pesticides and herbicides during the operation phase, particularly with the control of AIS. - Access should be limited to prevent further disturbance to the vegetation outside of the development footprint. - No fires should be allowed on site. - No plant or wood collection should be allowed outside of the proposed development footprint.
Increased stormwater runoff	- Prevent direct discharge of concentrated runoff into the Mputi River. - Stormwater infrastructure must be regularly inspected and maintained.
Risk of sewer pipeline leakage	- Sewer pipeline must be designed and installed to minimise the risk of leakage at the river crossing. - Regular inspection and maintenance must be undertaken. - Any leak or failure must be repaired immediately.
Ongoing disturbance of riparian habitat	- Avoid unnecessary clearing of riparian vegetation. - Disturbed areas must be rehabilitated after maintenance activities, and alien invasive plants should be controlled where they establish.

3. CLIMATE CHANGE ASSESSMENT

Climate change issues must be considered as part of the EIA process Please consider the Climate Change guideline. EAP must determine:

- The potential impact of climate change on society and the economy, whether the impact is negative or positive, considering that society needs to be at the centre of the proposed development;
- The potential alternatives of the proposed development, alternatives that will have less impact on climate change (environment and generation of waste included), the society and economy;
- whether, and to what extent, the proposed development will result in the release of greenhouse gas (GHG) emissions;
- whether the proposed development is necessary to achieve long term decarbonisation goals;
- the impact of the development on social, economic, natural and built environment that are crucial for climate change, adaptation and resilience;
- the projected impact of climate change on proposed development; and surrounding environment, and implications for the development.
- Explanation of how the impacts is likely to be exacerbated or minimised as result of climate change and what measures are likely to be implemented to accommodate and manage (adapt to) the anticipated worst scenario where applicable

h) whether, and to what extent, the impacts identified in (a) -(g) can be mitigated.

CLIMATE CHANGE ASSESSMENT

a) The potential impact of climate change on society and the economy, whether the impact is negative or positive, considering that society needs to be at the centre of the proposed development;

Climate change impacts:

Pre-construction

- Existing grassland vegetation contributes to the sequestration of atmospheric CO₂ through photosynthesis (+ve)
- Undisturbed soils store Soil Organic Carbon (SOC) and contribute to long-term carbon storage (+ve)
- Natural vegetation and permeable soils promote rainwater infiltration, groundwater recharge and reduced surface runoff (+ive)
- Existing natural drainage systems and vegetation provide climate resilience by reducing erosion and moderating local temperatures (+ve)

Construction phase:

- Site establishment, vegetation clearing and earthworks will result in the loss of vegetation and a temporary reduction in carbon sequestration (-ive)
- Disturbance of soils will release stored Soil Organic Carbon (SOC) and reduce the site's carbon storage capacity (-ive)
- Construction vehicles, machinery and generators will emit greenhouse gases (GHGs), contributing to the project's carbon footprint (-ive)
- The manufacture and transportation of construction materials will generate indirect greenhouse gas emissions (-ive)
- Increased dust generation and disturbed surfaces may reduce local air quality for nearby communities during construction (-ve)
- Temporary employment opportunities and local procurement will strengthen community resilience and adaptive capacity by supporting local livelihoods (+ve)

Rehabilitation (Post construction)

- Rehabilitation of disturbed areas with indigenous vegetation will restore ecosystem functions and promote the re-sequestration of CO₂ and Soil Organic Carbon (+ive)
- Stabilisation of exposed soils will reduce erosion and improve the site's resilience to intense rainfall events associated with climate change (+ve)
- Landscaping and tree planting around the conference centre will provide shade, improve the local microclimate and contribute to carbon sequestration (+ve)

Operational phases:

- Landscaping and maintained green spaces will continue to sequester CO₂, reduce the urban heat island effect and improve environmental quality (+ive)
- The incorporation of energy-efficient lighting, appliances and building systems will reduce electricity consumption and associated greenhouse gas emissions (+ive)
- The potential use of renewable energy technologies, such as solar photovoltaic systems, will reduce reliance on grid electricity and lower operational emissions (+ve)
- Water-efficient fixtures, rainwater harvesting (where feasible) and effective stormwater management will promote sustainable water use and improve resilience to drought conditions (+ve)
- Waste minimisation, recycling and the reuse of materials generated during operation will reduce resource consumption and associated climate impacts (+ve)
- Increased visitor and vehicle movements during conferences and events will result in greenhouse gas emissions from transport, although these impacts are expected to be localised and intermittent (-ve)
- Environmental awareness education of climate change (+ive)

- b) The potential alternatives of the proposed development, alternatives that will have less impact on climate change (environment and generation of waste included), the society and economy;**
 Alternatives considered include no-go, alternative site selection, and design alternatives. The no-go option would perpetuate the lack of local conferencing infrastructure, continued travel to neighbouring towns for conferences and events and limited economic development. Design alternatives with reduced climate impact include energy-efficient building design, use of solar photovoltaic systems, water-saving fixtures, and sustainable construction materials. These alternatives reduce energy demand, emissions, and waste generation while maintaining social and economic benefits.
- c) Determine whether, and to what extent, the proposed development will result in the release of greenhouse gas (GHG) emissions;**
 The proposed development will result in low to moderate GHG emissions, primarily during the construction phase due to material production, transport, and machinery use. During the operational phase, emissions will be limited and associated mainly with electricity consumption and waste generation. The incorporation of energy-efficient design measure and potential renewable energy use will significantly reduce long-term operational emissions.
- d) Determine whether the proposed development is necessary to achieve long term decarbonisation goals;**
 Decarbonisation will be achieved by incorporating low-carbon design measures such as energy-efficient systems, renewable energy, water conservation and waste minimisation.
- e) Determine the impact of the development on social, economic, natural and built environment that are crucial for climate change, adaptation and resilience;**
 The development will have **minimal negative impacts on the natural environment due to its location on a previously disturbed site**. Socially and economically, the project enhances adaptive capacity by improving access to a conference centre, local economic development and strengthening community resilience. The built environment will benefit from climate-responsive design, improving durability under changing climate conditions.
- f) Determine the projected impact of climate change on proposed development; and surrounding environment, and implications for the development.**
 Inter-annual changes in rainfall patterns, combination of droughts and floods, in the medium and long-term has negative impacts on structural integrity of facilities. The fluctuations in external forces (flows), severe floods, heating and cooling of structure cumulatively compromises the structures. Projected climate change impacts include higher temperatures, intense rainfall events, and periods of drought. These may affect building comfort, stormwater management, and water availability. Without mitigation, increased rainfall intensity could lead to localized erosion or flooding, while higher temperatures may increase cooling demands.
- g) Explanation of how the impacts is likely to be exacerbated or minimised as result of climate change and what measures are likely to be implemented to accommodate and manage (adapt to) the anticipated worst scenario where applicable**
 Climate change could be intensify risks such as stormwater runoff, heat stress, and water scarcity. These impacts will be minimised through appropriate stormwater management systems, passive cooling design, shade provision, water-efficient fittings, and durable construction materials. The design allows for future upgrades such as solar energy systems to further enhance resilience under worst-case climate scenarios.
- h) Determine whether, and to what extent, the impacts identified in (a) -(g) can be mitigated.**
 Climate considering engineering designs and on-going structural integrity inspections are necessary.

Taking the assessment of potential impacts into account, please provide an environmental impact statement that summarises the impact that the proposed activity and its alternatives may have on the environment after the management and mitigation of impacts have been taken into account, with specific reference to types of impact, duration of impacts, likelihood of potential impacts actually occurring and the significance of impacts.

Alternative A (preferred alternative)

The proposed development's environmental impacts are localised to the immediate receiving environment and are expected to remain manageable provided that the mitigation measures outlined in this BAR and EMP, as informed by specialist studies, are strictly implemented. The majority of identified negative impacts are anticipated to be short-term, localised, and of low to moderate significance, occurring mainly during the planning and construction phases to reduce the significance and severity of these impacts, all mitigation measures must be adhered to throughout the project lifecycle, with an ECO appointed to monitor compliance, particularly during construction, and an Operational Environmental Management Plan implemented during the operational phase to ensure continued environmental protection. Overall, the proposed development is expected to yield positive socio-economic benefits during both the construction and operational phases through the creation of temporary and permanent employment opportunities, thereby contributing to local economic development.

No-go alternative (compulsory)

The No-Go alternative has been evaluated and discredited. This alternative would avoid all construction-related environmental impacts. However, it would result in the continued shortage of educational infrastructure in the Aeroville and Somerset East area, representing a negative long-term social impact of moderate to high significance.

SECTION E. RECOMMENDATIONS OF PRACTITIONER

Is the information contained in this report and the documentation attached hereto sufficient to make a decision in respect of the activity applied for (in the view of the environmental assessment practitioner)?

YES	
YES	

Is an EMPr attached?

The EMPr must be attached as Appendix F.

If "NO", indicate the aspects that should be assessed further as part of a Scoping and EIA process before a decision can be made (list the aspects that require further assessment):

N/A

If "YES", please list any recommended conditions, including mitigation measures that should be considered for inclusion in any authorisation that may be granted by the competent authority in respect of the application:

PLANNING, DESIGN PHASE & PRE-CONSTRUCTION PHASE

- All relevant permits and authorisations including Water Use Licenses, Building Plan Approvals and plant removal permits must be in place prior to commencement of construction.
- It is recommended that an Environmental Control Officer be appointed to conduct independent audits and compile monthly audit reports to ensure compliance with the EMPr, EA and Water Use Licence during construction.
- Appropriate storm water structures must be designed to minimise erosion and sedimentation of watercourses.
- Rainwater tanks must be considered in the planning and design phase.
- The design and layout should consider the use of alternative energy sources.
- Conduct a floral and faunal search and rescue prior to construction commencement

CONSTRUCTION PHASE

- Necessary health and safety signage to be installed.
- No-go areas to be clearly and firmly barricaded using appropriate techniques
- Stock piles should be protected from the wind by covering them
- Rehabilitation of natural vegetation immediately after construction has finished, strictly in a systematic manner

REHABILITATION PHASE

- Re-vegetation of all disturbed and exposed surface with the same or similar indigenous vegetation at least 80% cover should be achieved in the construction site

OPERATION PHASE

- All relevant legislation and policy must be consulted, and the proponent must ensure that the project is compliant with such legislation and policy. The operational conditions outlined in the EA, WUL and EMP must be adhered to
- An Operational Environmental Management Plan to be compiled and implemented during operations to ensure continued environmental protection.
- Storm water management infrastructure must be properly maintained and monitored.
- Water-wise practices should be considered such as use of rainwater for non-portable water requirements.
- Effective monitoring and maintenance of all water reticulation infrastructures should be implemented to ensure there are no leakages.

SECTION F: APPENDICES

The following appendixes must be attached as appropriate:

Appendix A: Site plan(s)

Appendix B: Photographs

Appendix C: Facility illustration(s)

Appendix D: Specialist reports

Appendix E: Comments and responses report

Appendix F: Environmental Management Programme (EMPr)

Appendix G: Other information

